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Mohsin Ibna Hossain
AIUB, Physics 2 Final term

#Final TERM

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#PERIODIC MOTION (OSSCILATION)

Ch° 14

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Periodic Motion:-

⇒ A motion that repeats itself after equal intervals of time is known as periodic motion.

Example of periodic motion:-

1. A rocking chair,
2. A bouncing ball,
3. A tuning fork,
4. A swing in motion.
5. A water wave etc.

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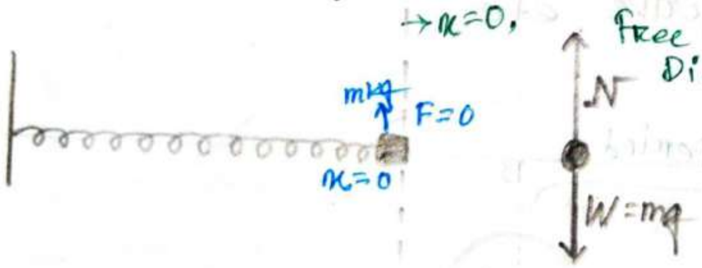
1. A body that undergoes the pericidic motion always has a stable equilibrium position.

2. When it moved away from the equilibrium position, a force comes into play to pull it back to the equilibrium position.

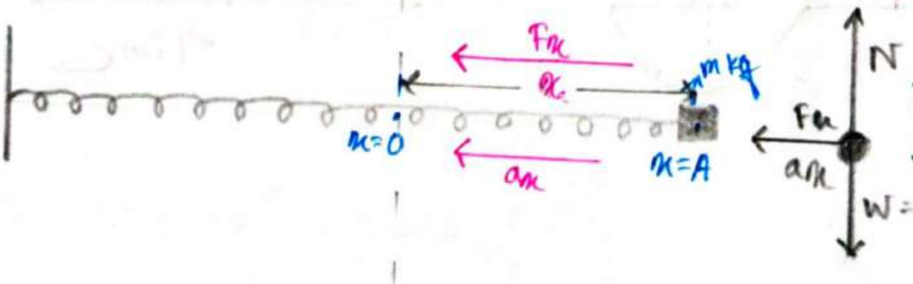
3. But the time it's gets there, it has picked up some kinetic energy, so it overshoots, stopping somewhere on the other side and again pulled back toward equilibrium.

(a)



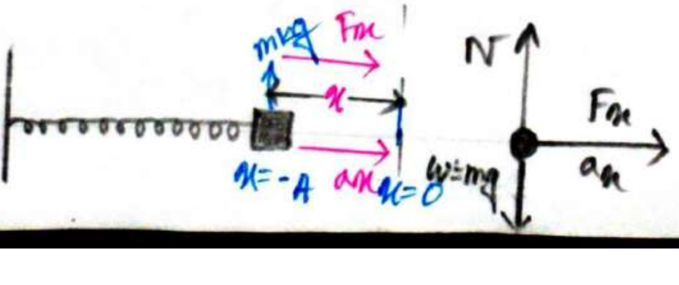
$x=0$; The relaxed spring exerts no force on the glider so acceleration = 0.

(b)



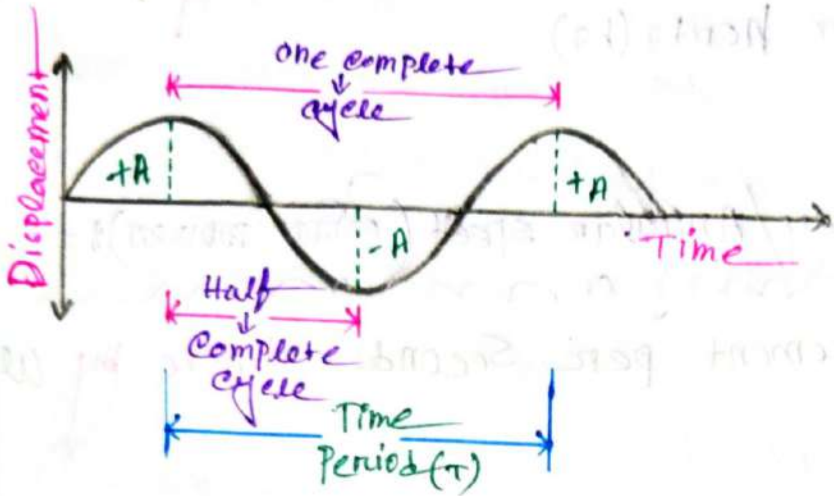
$x>0$; $F_x<0$, $a_x<0$; Stretched spring puls glider towards equilibrium position.

(c)



$x<0$; $F_x>0$, $a_x>0$; Compressed spring push glider towards equilibrium position.

Basic Quantities of periodic motion:



① Amplitude (বিস্তার):

⇒ The maximum displacement from the equilibrium position denote by A (it is always positive)

② Cycle:

⇒ One complete round trip (from A to -A and back to A through 0).

⇒ Half complete round trip (One side to another side that mean's A to -A)

③ Time period:

⇒ Time to complete one cycle. denote by T. unit (s)

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(4) Frequency:—

⇒ Number of cycle per second. denote by f.
unit (cycle/s) or hertz (hz)

(5) Angular Frequency/Angular speed (কৌণিক কম্পাঙ্ক):—

⇒ Angular displacement per second. denote by ω
unit (rad/s)

Relationship between Frequency & Time period

⇒ $\boxed{f = \frac{1}{T}}$

Relationship between Angular Frequency & Frequency & Time Period

⇒ $\boxed{\omega = 2\pi f = 2\pi \times \frac{1}{T}}$

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#Check point#

Problems 14.1

⇒ An ultrasonic transducer used for medical diagnosis oscillates at $6.7\ MHz = 6.7\times10^6\ Hz$. How long does each oscillation take, and what is the angular frequency.

⇒ In given:—

$$f = 6.7\times10^6\ Hz$$

(i) We know:—

$$f = \frac{1}{T}$$

$$\therefore T = \frac{1}{f}$$

$$= \frac{1}{6.7\times10^6} = 1.49\times10^{-7}\ sec$$

(ii) We know:—

$$\omega = 2\pi f = 2\pi\times\frac{1}{T}$$

$$= 2\pi\times6.7\times10^6$$

$$= 4.20\times10^7\ rad/s$$

#Wing Frequencies:

1. Hummingbird = 80 Hz
2. Insects (পোকামাকড়) = 330 Hz
3. House fly (মাছি) = 600 Hz
4. Mosquito (মশা) = 1040 Hz

#Simple Harmonic Motion (SHM)

⇒ SHM is the special kind of periodic motion in which the the restoring force is directly proportional to the displacement. This is happen if the spring is an ideal one.

We know:

$$F_x = -Kx \qquad [K = \text{Spring constant}]$$

$$\therefore ma_x = -Kx$$

$$a_x = -\frac{K}{m} \times x \qquad \left[\omega^2 = \frac{K}{m} = \text{constant}\right]$$

$$a_x = -\omega^2 x$$

$$a_x \propto -x$$

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Note:- ① The minus sign means acceleration and displacement always have opposite signs.

② A body that undergoes SHM in called harmonic oscillator.

Exploring the Diversity of periodic motion and SHM:-

①. An SHM motion is always periodic motion. But A periodic motion may or may not be SHM.

②. If the amplitude in small enough the oscillations of a periodic system are approximately SHM.

③. The SHM/oscillatory motion refers to the motion in which the object moves back and forth repeatedly while periodic motion refers to the motion in which the objects repeats a path after a regular interval of time.

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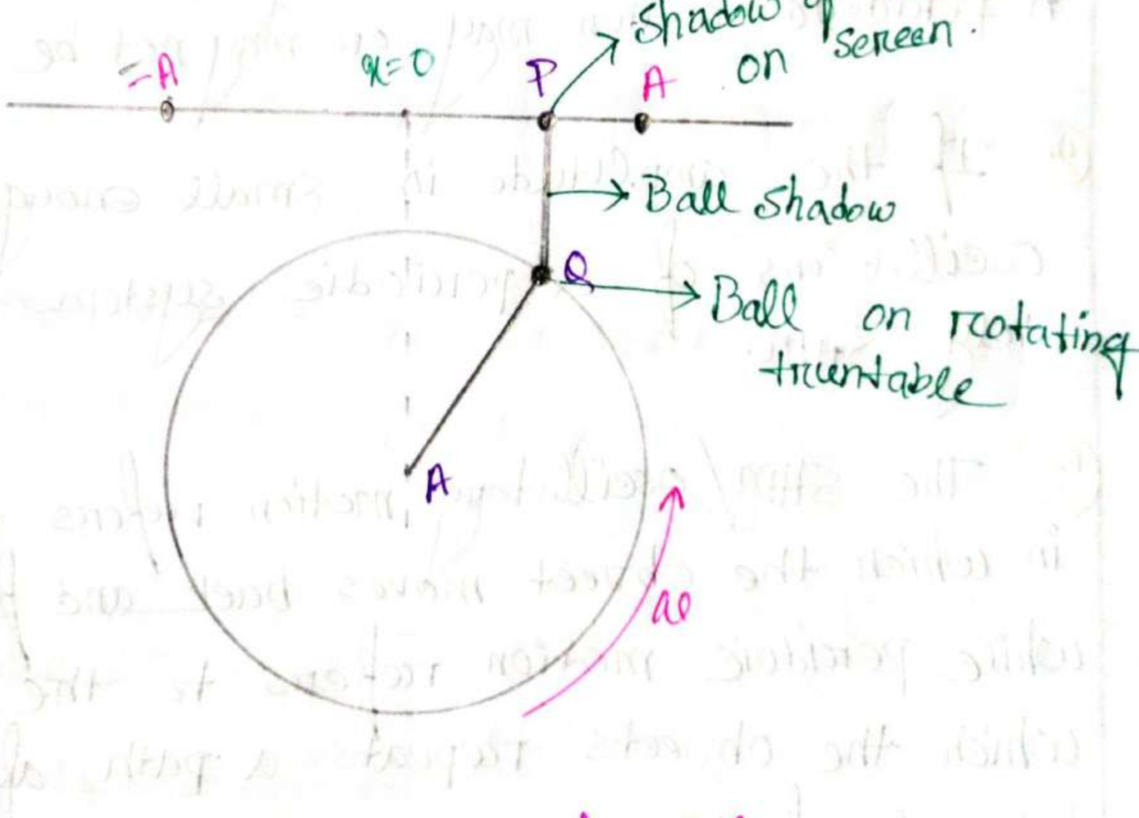
(4) All oscillatory motions are periodic but All periodic motion are not oscillatory.

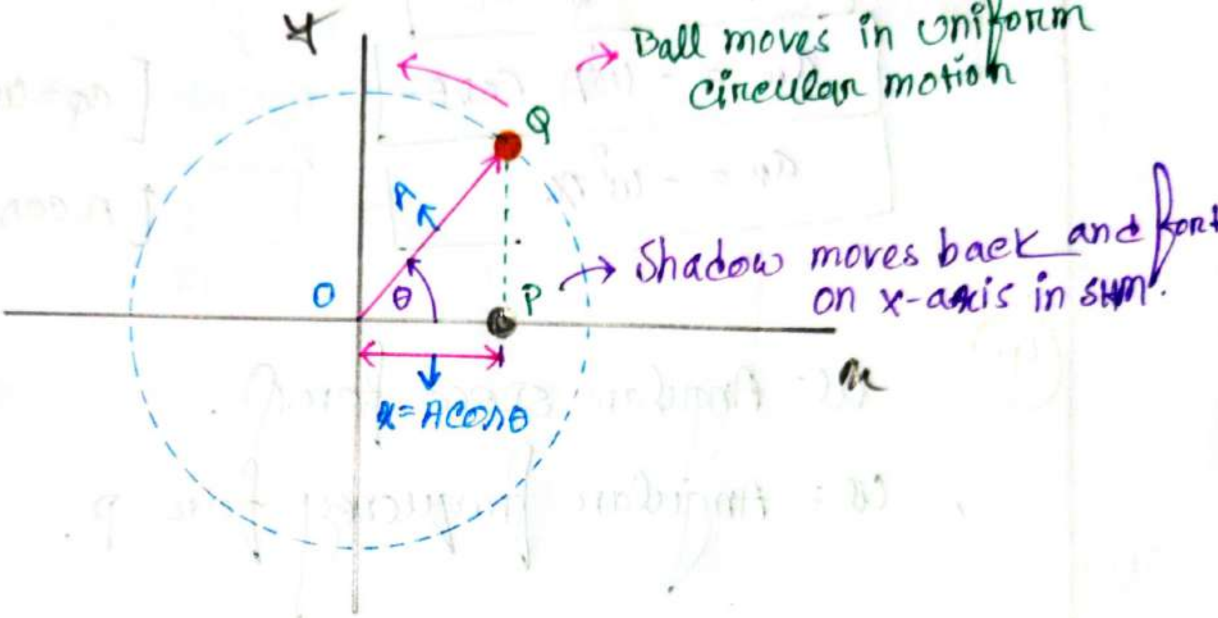
Uniform Circular Motion and Equation of SHM:—

(a) Relating Uniform circular motion and simple harmonic motion:—



Note:— SHM is the projection of uniform circular motion onto the diameter.

(b) The ball's shadow moves exactly like a body Oscillating



(1) Displacement of the point P : $\boxed{x = A\cos\theta}$

(2) Centripetal Acceleration (কেন্দ্রমুখী ত্বরণ) of the ball Q :—

$\therefore \boxed{a_Q = \omega^2 A}$

$\omega = \dfrac{\text{Angular displacement}}{\text{time duration.}}$

ω = Angular speed

A = radius of the circle.

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(3) Acceleration of the projection point P:—

$$\boxed{a_x = -a_Q \cos\theta}$$
$$\boxed{a_x = -\omega^2 A \cos\theta}$$
$$\boxed{a_x = -\omega^2 x}$$

$[a_Q = \omega^2 A]$

$[A\cos\theta = x]$

(4.) ω = Angular speed for Q.
ω = Angular frequency for P.

(5.) If point Q makes one complete revolution in time T,
Then P goes through one complete cycle of oscillation in the same time T.

So, we get:—

$$\boxed{\omega = \frac{2\pi}{T} = 2\pi f}$$

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Equations of SHM:—

(1) Angular Frequency of a harmonic oscillator of mass m and spring constant K :-

$$\boxed{\omega = \sqrt{\frac{K}{m}}}$$

ω = Angular frequency
K = spring constant
m = mass.

(2) Frequency :-

$$\boxed{f = \frac{\omega}{2\pi}}$$

$$\boxed{f = \frac{1}{2\pi} \times \sqrt{\frac{K}{m}}}$$

f = frequency

(3) Time period:—

$$\boxed{T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi \times \sqrt{\frac{m}{K}}}$$

T = time period

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#Check point:—

⇒ Considere a harmonic with mass 50g and force Constant 2000 N/m. Calculate the

(i) Angular frequency.

(ii) Frequency and

(iii) Time period. of the oscillator.

If you change the mass by 100g what would be the

(iv) Angular frequency

(v) Frequency and

(vi) Time period of the osillator.

If the force Constant change by 1000 N/m for 50g oscillator what would be the

(vii) Angular frequency

(viii) Frequency and

(ix) Time period of the osillator.?

Compare the results and make your comments.

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⇒ In given:—

$m = 50 \times 10^{-3}$ kg

$k = 2000$ N/m

(i)

We know:—

$$\omega = \sqrt{\frac{k}{m}}$$

$$= \sqrt{\frac{2000}{50 \times 10^{-3}}}$$

$$= 200 \text{ rad/s}$$

(ii)

We know:—

$$f = \frac{\omega}{2\pi}$$

$$= \frac{200}{2 \times 3.1416}$$

$$= 31.83 \text{ Hz}$$

(iii)

We know:—

$$T = \frac{1}{f}$$

$$= \frac{1}{31.83} = 0.031 \text{ sec}$$

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Now—

Is given—

$m = 100 \times 10^{-3}$ kg

$k = 2000$ N/m

(iv)

We know—

$$\omega = \sqrt{\frac{k}{m}}$$

$$= \sqrt{\frac{2000}{100 \times 10^{-3}}}$$

$$= 141.42 \text{ rad s}^{-1}$$

(v)

We know—

$$f = \frac{\omega}{2\pi}$$

$$= \frac{141.42}{2 \times 3.1416}$$

$$= 22.50 \text{ Hz}$$

(vi)

We know—

$$T = \frac{1}{F}$$

$$= \frac{1}{22.50} = 0.04 \text{ sec}$$

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NOW:— Is given:—

$m = 50 \times 10^{-3}$ kg

$K = 1000\ Nm^{-1}$

(vii)

We know:—

$$\omega = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{1000}{50 \times 10^{-3}}}$$

$$= 141.42 \text{ rad/s}$$

(viii)

We know:—

$$f = \frac{\omega}{2\pi}$$

$$= \frac{141.42}{2 \times 3.1416}$$

$$= 22.50 \text{ Hz}$$

(ix)

We know:—

$$T = \frac{1}{f} = \frac{1}{22.50} = 0.044 \text{ sec}$$

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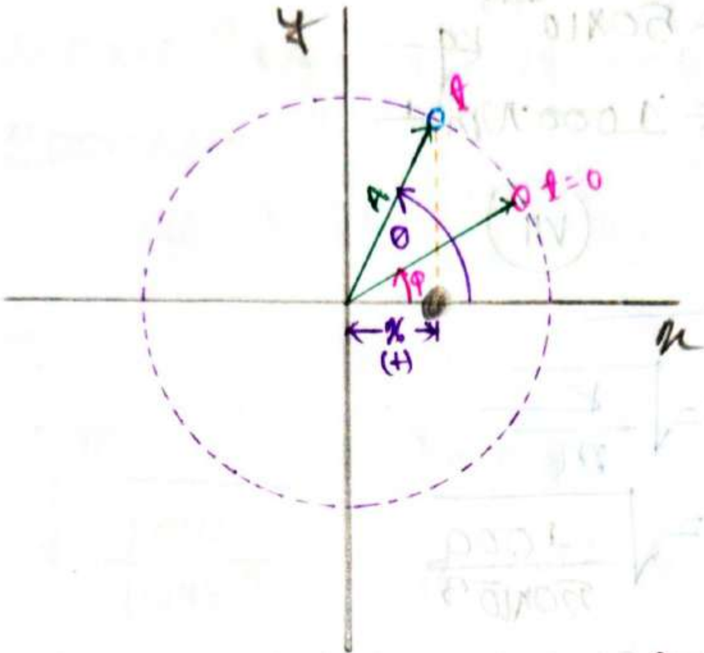
Displacement in SHM:-



① Angular Displacement is mesured from (Positive) x-axis in the Counter/Anti-clockwise direction.

② At the starting time $t=0$,
Angular displacement from x-axis $= \phi$

③ At time $= t$
Angular displacement from x-axis $= \theta$

(3+1) ∴ Angular speed $\omega = \dfrac{\theta - \phi}{t}$

$\therefore \theta = \omega t + \phi$

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(ग) We know:—

Displacement of the Oscillators—

$x = A\cos\theta$

$$\boxed{x = A\cos(\omega t + \phi)}$$

A = Amplitude (বিস্তার)

ω = Angular Frequency / Angular speed (কৌণিক কম্পাঙ্ক)

ϕ = Phase Angle (আদি দশা)

$\theta = (\omega t + \phi)$ = Phase Angle After time t (t সময় পর দশা কোণ)

Velocity and Accelaration In SHM

(1) Velocity $V = \frac{dx}{dt} = \frac{d}{dt} A\cos(t\omega + \phi)$

$$\therefore \boxed{V = -A\omega\sin(\omega t + \phi)}$$

(1.1)

$$\boxed{\begin{aligned} V_{max} &= A\omega \\ -V_{max} &= -A\omega \end{aligned}}$$

$\therefore$ Velocity Oscilates between $V_{max} = A\omega$ to $-V_{max} = -A\omega$

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(2) Acceleration $a = \frac{dv}{dt} = \frac{d}{dt}\left(-\omega A \sin(\omega t + \phi)\right)$

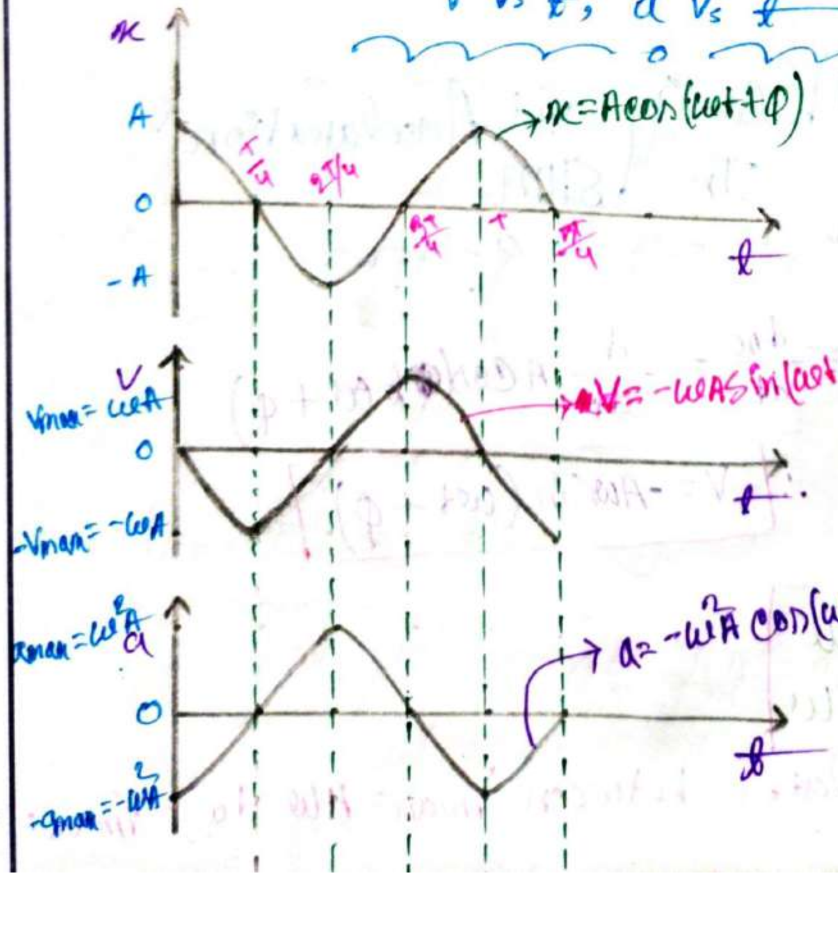
$\therefore \boxed{a = -A\omega^2 \cos(\omega t + \phi)}$ → In terms of time

or $\boxed{a = -\omega^2 x}$ → In terms of Displacement

(2.1) $\boxed{a_{max} = A\omega^2}$

⇒ Acceleration a oscillates between $a_{max} = A\omega^2$ to $-a_{max} = -A\omega^2$

Graph of x vs t, V vs t, a vs t



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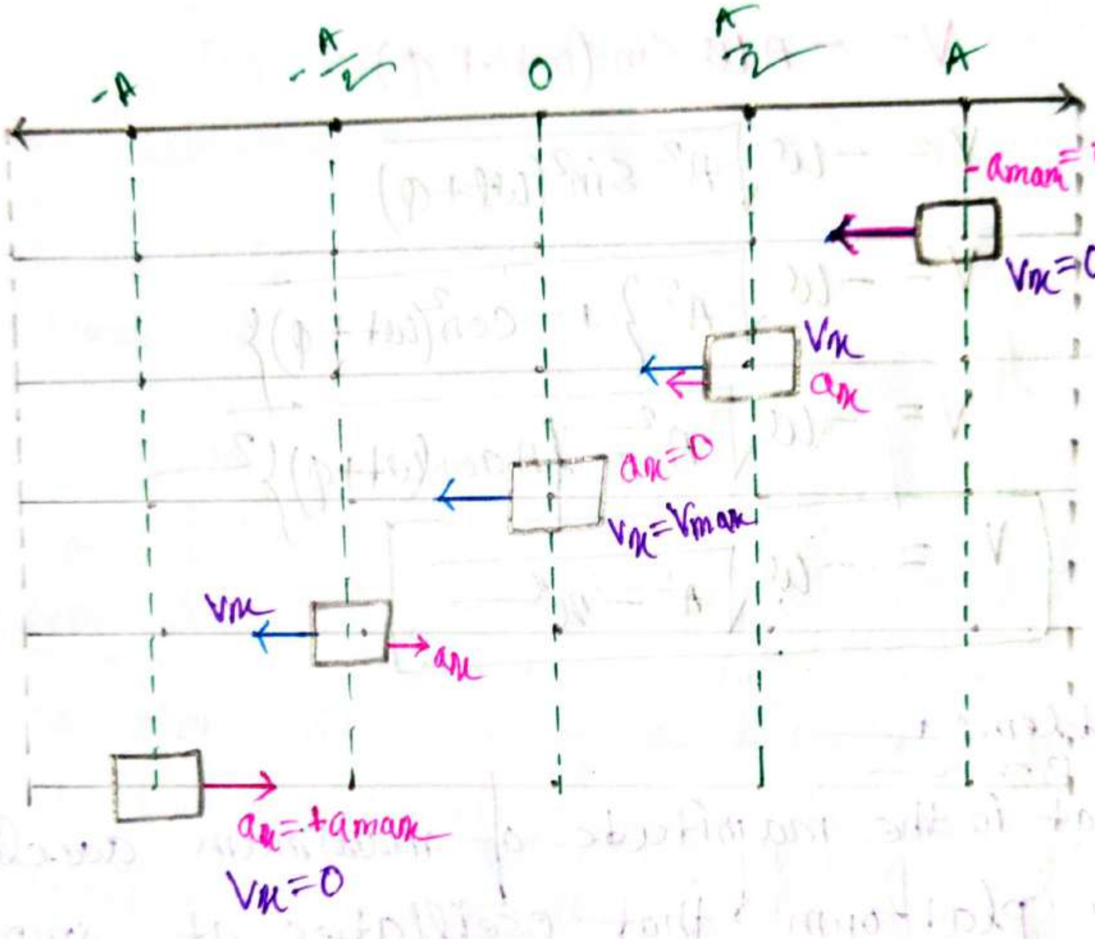
How Velocity and acceleration Vary During One cycle:—



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Velocity In terms of Displacement :—

We know :—

$$V = -A\omega \sin(\omega t + \phi)$$

$$V = -\omega\sqrt{A^2 \sin^2(\omega t + \phi)}$$

$$V = -\omega\sqrt{A^2\{1 - \cos^2(\omega t + \phi)\}}$$

$$V = -\omega\sqrt{A^2 - \{A\cos(\omega t + \phi)\}^2}$$

$$\boxed{V = -\omega\sqrt{A^2 - x^2}}$$

problem : 1

⇒ What is the magnitude of maximum acceleration of a platform that oscillates at amplitude 2.20cm and frequency 6.60 Hz ?

⇒ In given :—

$A = 2.20\text{cm} = 2.20 \times 10^{-2}\text{m}$

$f = 6.60\text{Hz}$

We know :—

maximum acceleration $a_{max} = A\omega^2$ —— (1)

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So, we know,—

$$\omega = 2\pi f$$

$$= 2 \times 3.1416 \times 6.60$$

$$= 41.46 \ rad\,s^{-1}$$

Now, Substitute the value of ω in equation (1) ⇒

$$a_{max} = \omega^2 A$$

$$= (41.46)^2 \times 2.20 \times 10^{-2}$$

$$= 37.8 \ m s^{-2}$$

Problem: 2:—

⇒ An Oscillator consists of a block of mass 0.5kg Connected to a spring. When set into oscillation with amplitude 35.0cm the oscillator repeats it's motion every 0.5sec. Find the

(a) Time period

(b) frequency

(c) Angular frequency

(d) spring constant

(e) maximum speed

(f) magnitude of the maximum force on the block from the spring

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⇒ Is given:—

$m = 0.5 kg$

$A = 35 \times 10^{-2} m$

$T = 0.5 Sec$

(a)

Time period $T = 0.5 Sec$

(b)

We know:—

$f = \frac{1}{T}$

$= \frac{1}{0.5} = 2Hz$

(c)

We know:—

$\omega = 2\pi f$

$= 2 \times 3.1416 \times 2$

$= 12.56 \ rad \ s^{-1}$

(d)

We know:—

$\omega = \sqrt{\frac{K}{m}}$

$\therefore K = \omega^2 m$

$= (12.56)^2 \times 0.5$

$= (78.88) N/m$

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(e)

We know:—

$V_{max} = A\omega$

$= 35 \times 10^{-2} \times (12.56)$

$= 4.39\ ms^{-1}$

(F)

We know:—

$F_{max} = m a_{max}$ ------ (1)

$a_{max} = A\omega^2$

$= 35 \times 10^{-2} \times (12.56)^2$

$= 55.21\ ms^{-2}$

Substitute the value of a_{max} in equation (1) ⇒

$\therefore F_{max} = m a_{max}$

$= 0.5 \times 55.21$

$= 27.605\ N$

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Energy in SHM:-

⇒ The force exerted by an ideal spring is a conservative force, so the total mechanical energy is conserved.

$$\boxed{E = K + U = \text{Constant}}$$

E = Mechanical Energy

K = Kinetic Energy

U = potential Energy

Potential Energy:-

⇒ The potential Energy of a linear oscillator is associated entirely with the spring. It's value depends on the how much the spring is stretched or compressed, that is on. $x(t)$:

We know:-

$$\boxed{U = \frac{1}{2}kx^2} \rightarrow \text{In terms of Displacement}$$

$$\boxed{U = \frac{1}{2}kA^2\cos^2(\omega t + \phi)} \rightarrow \text{In terms of time.}$$

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Kinetic Energy:—

3) The kinetic Energy of the system in associated entirely with the block. It's value depends on how fast the block is moving, that is on $V(t)$

We know:—

$$K = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m\{-\omega A\sin(\omega t+\phi)\}^2$$

$$= \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t+\phi)$$

$$\boxed{K = \frac{1}{2}KA^2\sin^2(\omega t+\phi)}$$ → In terms of time.

We know:—

$$V = -\omega\sqrt{A^2-x^2}$$

$$K = \frac{1}{2}mv^2$$

$$= \frac{1}{2}m\omega^2(A^2-x^2)$$

$$= \frac{1}{2}m \times \frac{K}{m}(A^2-x^2)$$

$$\boxed{K = \frac{1}{2}K(A^2-x^2)}$$ → In terms of Displacement

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Mechanical Energy:-

We know-

$$E = U + K$$

$$= \frac{1}{2}KA^2\cos^2(\omega t+\phi) + \frac{1}{2}KA^2\sin^2(\omega t+\phi)$$

$$= \frac{1}{2}KA^2\left(\cos^2(\omega t+\phi) + \sin^2(\omega t+\phi)\right)$$

$$= \frac{1}{2}KA^2 \times 1$$

$$\boxed{E = \frac{1}{2}KA^2}$$

Note:- The mechanical energy of a linear oscillator is indeed constant and independent of Time.

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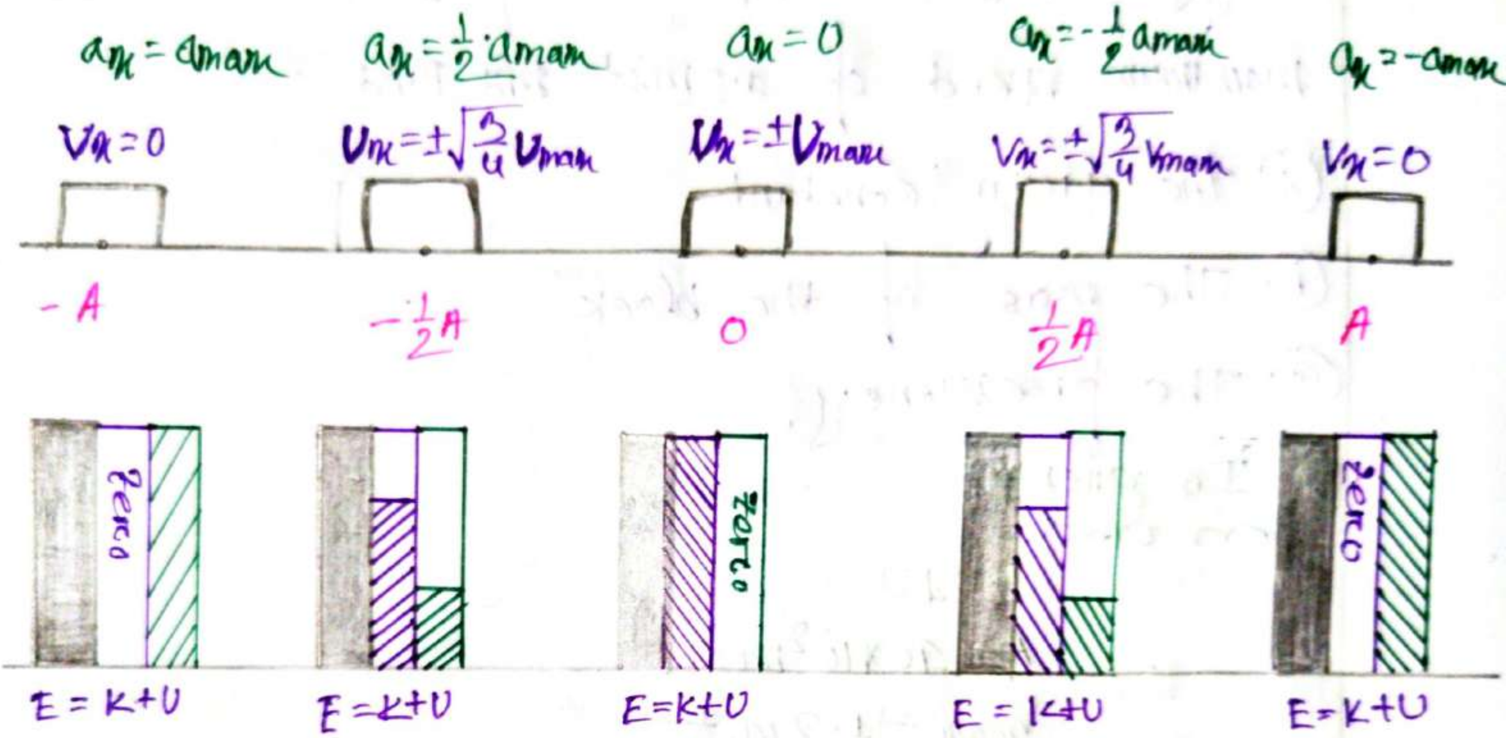
Graph of E, K and U Versus Displacement in SHM



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Problem: 3

⇒ An oscillating block-spring system has a mechanical energy of 1 J an amplitude of 10 cm. and a maximum speed of $1.2 ms^{-1}$. Find

(a) The spring constant

(b) The mass of the block

(c) The frequency.

⇒ In given:—

$$E = 1 J$$

$$A = 10 \times 10^{-2} m$$

$$V_{max} = 1.2 ms^{-1}$$

(a)

We know:—

$$\omega = \sqrt{\frac{k}{m}}$$

$$E = \frac{1}{2} kA^2$$

$$2E = kA^2$$

$$\therefore k = \frac{2E}{A^2}$$

$$= \frac{2 \times 1}{(10 \times 10^{-2})^2}$$

$$= 200 N/m$$

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(b)

We know:—

$$\omega = \sqrt{\frac{K}{m}}$$

$$\omega^2 = \frac{K}{m}$$

$$\therefore m = \frac{K}{\omega^2}$$

$$= \frac{200}{(12)^2}$$

$$= 1.388 \text{ Kg}$$

We know:—

$$V_{max} = \omega A$$

$$\therefore \omega = \frac{V_{max}}{A}$$

$$= \frac{1.2}{10 \times 10^{-2}}$$

$$= 12$$

(c)

We know:—

$$f = \frac{\omega}{2\pi}$$

$$= \frac{12}{2 \times 3.1416}$$

$$= 1.9098 \text{ Hz}$$

(d)

We know:—

$$T = \frac{1}{f} = \frac{1}{1.9098} = 0.5 \text{ sec}$$

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Problem 4:—

⇒ A 5.00kg on a object on a horizontal frictionless Surface is attached to a spring with $k = 1000$ N/m. The object in displaced from equilibrium 50.cm horizontally and given an initial velocity of $10ms^{-1}$ back toward the equilibrium position. What are

(a) The motion frequency.

(b) The initial potential energy.

(c) The initial kinetic energy.

(d) The motion's Amplitude.

⇒ in given:—

$k = 1000 Nm^{-1}$ | $m = 5kg$

$x_i = 50 \times 10^{-2} m$

$V_i = 10ms^{-1}$

(a)

We know:—

$f = \frac{\omega}{2\pi}$

$= \frac{10\sqrt{2}}{2\pi}$

$= 2.25 Hz$

We know:—

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{1000}{5}}$

$= 10\sqrt{2} \ rad s^{-1}$

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(b.)

We know:—

$$U = \frac{1}{2}kx^2$$

$$= \frac{1}{2} \times 1000 \times (50 \times 10^{-2})^2$$

$$= 125 J.$$

(c)

We know:—

$$K = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 5 \times (10)^2$$

$$= 250 J$$

(d.)

We know:—

$$E = \frac{1}{2}kA^2$$

$$\therefore A = \sqrt{\frac{2E}{k}}$$

$$= \sqrt{\frac{2 \times 375}{1000}}$$

$$= 0.8660 m$$

We know:—

$$E = U + K$$

$$= 125 + 250$$

$$= 375 J$$

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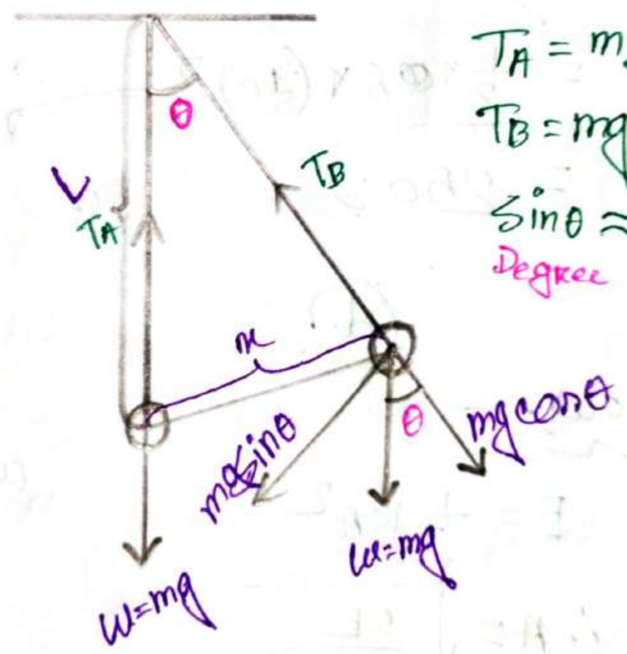
Application of SHM:-

⇒ Here we will discuss two application:-

1. Simple Pendulum
2. Physical Pendulum.

Simple Pendulum:-



$T_A = mg$

$T_B = mg\cos\theta$

$\sin\theta \approx \theta$

Degree Radian

1. Restoring force $F_\theta = -mg\sin\theta \approx -mg\theta$

$$\therefore F_\theta = -mg\frac{x}{L} = -\frac{mg}{L}x$$

2. Comparing with Hooke's law ($F = -kx$)

$$\therefore k = \frac{mg}{L}$$

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3. Angular Frequency :—

$$\omega = \sqrt{\frac{k}{m}}$$

$$= \sqrt{\frac{mg}{mL}}$$

$$\boxed{\omega = \sqrt{\frac{g}{L}}}$$

4. Frequency :—

$$f = \frac{\omega}{2\pi}$$

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}}$$

5. Time period :—

$$T = \frac{1}{f}$$

$$\boxed{T = 2\pi\sqrt{\frac{L}{g}}}$$

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Check point:—

Problem:- 14.8:—

⇒ Find the period and frequency of a simple pendulum 1m long at a location when $g = 9.8 ms^{-2}$

⇒ Is given:—

$L = 1m$

$g = 9.8 ms^{-2}$

We know:—

(i) $f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$

$= \frac{1}{2\pi} \times \sqrt{\frac{9.8}{1}}$

$= 0.49 Hz$

(ii) $T = 2\pi\sqrt{\frac{L}{g}}$

$= 2\pi \times \sqrt{\frac{1}{9.8}}$

$= 2.007 sec$

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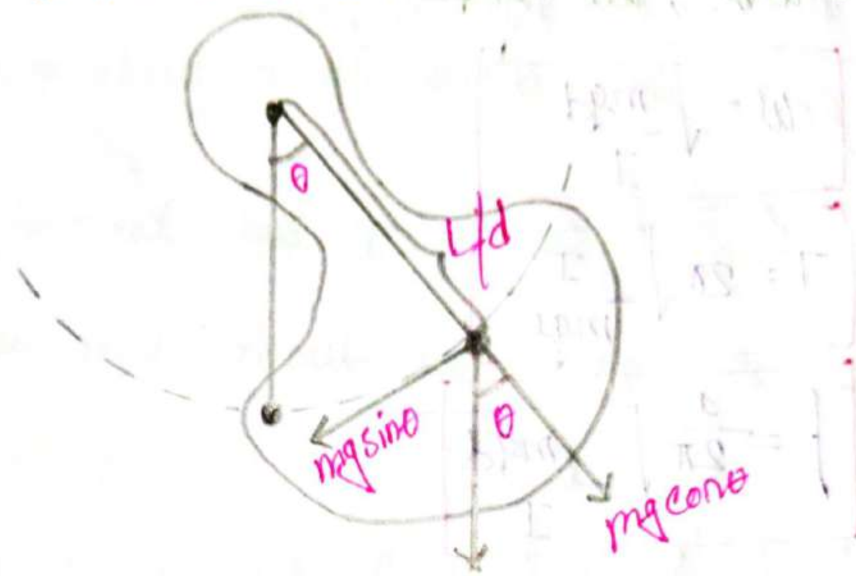
Physical Pendulum:-



⇒ For small oscillations, mg analyzing the motion of real physical pendulum is almost as easy as for a simple pendulum.

(1) The weight mg causes a restoring torque :-

$$\tau = -(mg)(d\sin\theta)$$
$$= -(mgd)\theta$$

(2) According to Newton's 2nd Law :-

$$F = I\alpha$$
$$-(mgd)\theta = I\alpha$$
$$-(mgd)\theta = I\frac{d^2\theta}{dt^2}$$
$$\frac{d^2\theta}{dt^2} = -\frac{mgd}{I}\theta$$

I = moment of inertia
α = Angular acceleration.

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(8.) Comparing with acceleration equation of SHM $a = -\omega^2 x$, we get

$$\boxed{\omega = \sqrt{\frac{mgd}{I}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{mgd}}}$$

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{mgd}{I}}}$$

Common Moments of Inertia :-

(1.) Solid cylinder or disc, symmetry axis / central axis $I = \frac{1}{2}mR^2$

where, m = mass

R = radius

(2.) A Hoop or A ring, central axis/symmetry axis $I = mR^2$

(3.) Solid sphere, central axis $I = \frac{2}{5}mR^2$

(4.) A Uniform rod, about center $I = \frac{1}{12}mL^2$

L = Length

m = mass

(5.) A cylinder about central diameter $I = \frac{1}{4}mR^2 + \frac{1}{12}mL^2$

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(6) A solid cylinder about End diameter $I = \frac{1}{4}mR^2 + \frac{1}{3}mL^2$

(7) A Thin hoop about a diameter $I = \frac{1}{2}mR^2$

(8) A Thin spherical shell $I = \frac{2}{3}mR^2$

(9) A uniform rod about the end $I = \frac{1}{3}mL^2$

problem no: 5:—

⇒ A uniform rod with length L, provoted at one end. what in the period of its motion as a pendulum? Considering the rod as a meter stick calculate the time period. Take $g = 9.8\,ms^{-2}$ and length $L = 1m$

⇒ In given:—

$g = 9.8\,ms^{-2}$
$L = 1m$
$d = \frac{L}{2} = \frac{1}{2}$

We know:—

The movement of inertia for a uniform rod about the end

$I = \frac{1}{3}mL^2$

Time pediod:—

$$T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{\frac{1}{3}mL^2}{mgd}}$$

$$= 2\pi\sqrt{\frac{L^2}{3gd}} = 2\pi\sqrt{\frac{1^2}{3\times 9.8\times\frac{1}{2}}} = 1.6387\,\text{Sec}$$

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(b) What is the Angular frequency ?

(c) What is the Frequency ?

⇒ We know:-

(b)

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$= \sqrt{\frac{mg \times \frac{1}{2}}{\frac{1}{3}m(1)^2}}$$

$$= \sqrt{\frac{3 \times 9.8}{1^2 \times 2}}$$

$$= 3.83 \text{ rad s}^{-1}$$

We know:-

$$I = \frac{1}{3}ml^2$$

(c)

We know:-

$$f = \frac{1}{2\pi} \times \sqrt{\frac{mgd}{I}}$$

$$= \frac{1}{2\pi} \times 3.83$$

$$= 0.6095 \text{ Hz}$$

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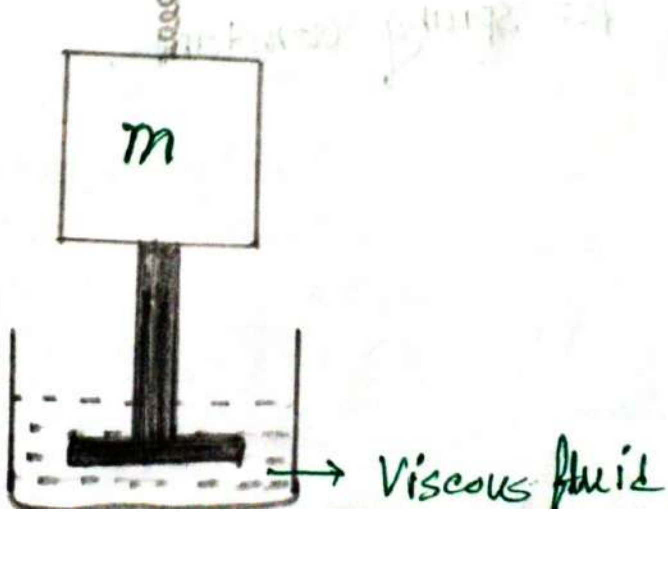
#Damped Oscillation:-

⇒ We know:-

⇒ when we swing a pendulum, it will eventually come to rest due to air friction and pressure at the support, this motion is Damped Oscillation.

⇒ Damped Oscillation is a type of periodic motion where an oscillator's amplitude gradually reduces due to frictional forces. The energy dissipates continuously, and for small damping, the motion remains approximately periodic before eventually coming to rest.



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⇒ The damping force depends on the nature of the surrounding medium.

⇒ Damping force (F_d) is proportional to the velocity (v) of the oscillator and acts opposite to the direction of velocity

$$\therefore F_d \propto -v$$

$$F_d = -bv \quad \text{——} \quad (i)$$

b = Damping coefficient

⇒ When the block is displaced from the equilibrium position ($x=0$). A restoring force (F_s) from the spring acts opposite direction of the displacement (x).

$$F_s = -kx \quad \text{——} \quad (ii)$$

k = spring constant.

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so, total force acting on the mass at any time:—

$$F_{net} = F = -kx + (-bv)$$

$$ma = -kx - bv$$

$$m\left(\frac{d^2x}{dt^2}\right) = -kx - b\left(\frac{dx}{dt}\right)$$

⇒ This is differential equation of x. and its Solution gives the displacement for damped oscillator.

(1) ⇒ For a small damping force, the displacement equation Canbe found as:—

$$\boxed{x = \left[Ae^{-\left(\frac{b}{2m}\right)t}\right] \cos(\omega' t + \phi)}$$

(2) Amplitude (A'):—

$$\boxed{A' = Ae^{-\left(\frac{b}{2m}\right)t}}$$

(3) Angular Freqency (ω'):—

$$\boxed{\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}}$$

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A = initial Amplitude.

$\beta = \frac{b}{2m}$ = damping factor.

ω' = natural Angular frequency

ϕ = Phase constant.

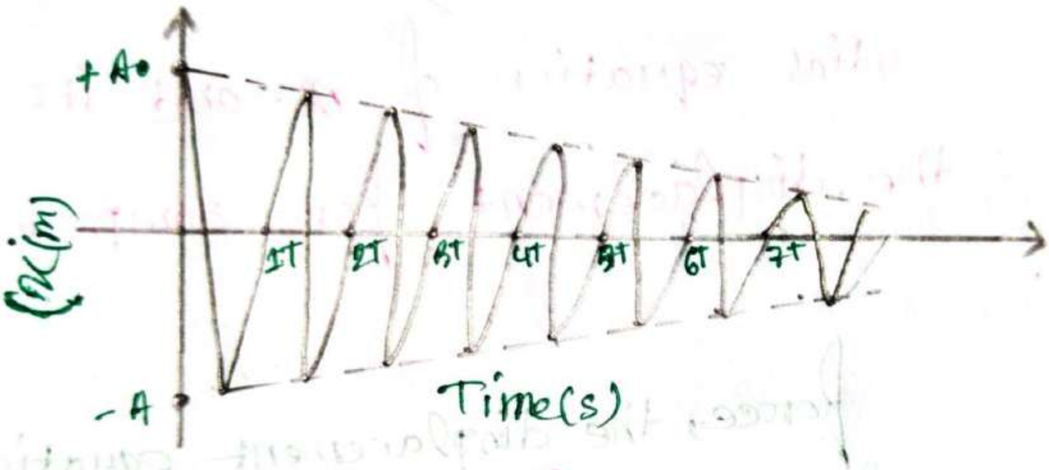

Fig:- The amplitude decreases exponentially with time.

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Mechanical Energy of Damped Oscillation :-

We know :-

$$E = \frac{1}{2} KA'^2$$

$$= \frac{1}{2} K \left[A e^{-\left(\frac{b}{2m}\right) t} \right]^2$$

$$\boxed{E = \frac{1}{2} KA^2 e^{-\left(\frac{b}{m}\right) t}}$$

Note :- The mechanical energy of damped oscillation is not constant but decreases continuously, Approaching zero after a long time. Because damping force is nonconservative.

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problem:-06:-

⇒ For the damped oscillator system shown in Fig, with $m = 250g$, $k = 85 N/m$, and $b = 70 g/s$, what is the ratio of the oscillation amplitude at the end of 20 cycles to the initial oscillation amplitude?

⇒ In given;-

$m = 250g = 250 \times 10^{-3} kg$

$k = 85 Nm^{-1}$

$b = 70 g/s = 70 \times 10^{-3} kg/s$

Let, Time period $= T'$

So, Time for 20 cycles $t = 20T'$

We know:-

At $t = 20T'$

$$A' = Ae^{-\left(\frac{b}{2m}\right)t}$$

$$= Ae^{-\left(\frac{70\times10^{-3}}{2\times250\times10^{-3}}\right)20\times0.34}$$

$$= A \times 0.3859$$

We know:-

$$\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)}$$

$$= \sqrt{\frac{85}{250\times10^{-3}} - \left(\frac{70\times10^{-3}}{2\times250\times10^{-3}}\right)}$$

$$= 18.43 \text{ rad/s}$$

$$T' = \frac{2\pi}{\omega'} = \frac{2\pi}{18.43}$$

$$= 0.34 s$$

At $t = 0$,

$$A' = Ae^{-\left(\frac{b}{2m}\right)t}$$

$$= Ae^{-0}$$

$$= A \times 1$$

$$= A$$

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So, the ratio of amplitudes $= \frac{A'}{A''}$

$$= \frac{A \times 0.3859}{A}$$

$$= 0.3859$$

Problems for Practice

Problem 14.3:—

⇒ The tip of tuning fork goes through 440 complete vibration in 0.500 sec. Find

(a) The time period of the motion.

(b) The Angular frequency.

⇒ In given:—

$$N = 440$$

$$\text{time taken } (t) = 0.500 \text{ sec}$$

(a)

We know:—

$$T = \frac{t}{N}$$

$$= \frac{0.500}{440}$$

$$= 1.136 \times 10^{-3} \text{ sec}$$

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(b)

We know:—

$\omega = 2\pi f$

$\omega = 2\pi \times \frac{1}{T}$

$= 2 \times 3.1416 \times \frac{1}{1.136 \times 10^{-3}}$

$= 5522.20$ rad/s

Problem: 14.4:—

⇒ The displacement of an oscillating object as a function of time is shown in Fig. E14.4. What are

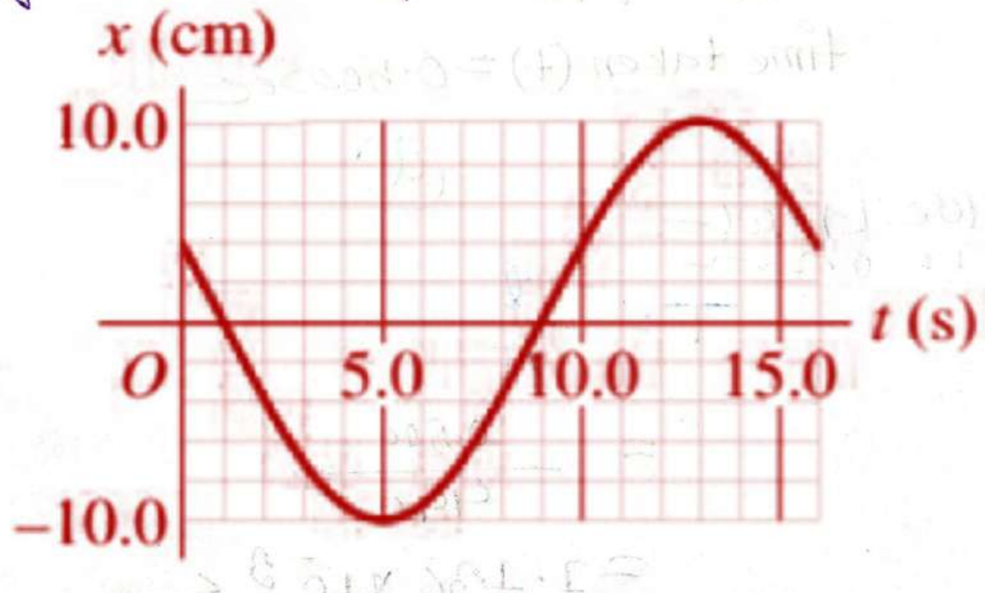
(a) The frequency.

(b) The amplitude

(c) The time period.

(d) The angular frequency.

Figure **E14.4**



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In given:—

$T = 16$ sec

$A = 10\,cm = 10 \times 10^{-2} m$

(a)

We know:—

$$f = \frac{1}{T}$$

$$= \frac{1}{16} = 0.0625\,Hz$$

(b)

$A = 10\,cm = 10 \times 10^{-2} m$

(c)

$T = 16$ Sec

(d)

We know:—

$$\omega = 2\pi f$$

$$= 2\pi \times 0.0625$$

$$= 0.3926 \text{ rad/s}$$

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Problem : 14.5:—

⇒ A machine part is undergoing SHM with a frequency of 5Hz and amplitude 1.80 cm. How long does it take the part to go from $x=0$ to $x=-1.80$ cm.

⇒ It's given:—

$$f = 5Hz$$

$$A = 1.80cm$$

time taken to move from $x=0$ to $x=-1.80cm$

We know:—

$$f = \frac{1}{T}$$

$$\therefore T = \frac{1}{f} = \frac{1}{5} = 0.2 \text{ sec}$$

T = Time period for one complete cycle.

But for SHM to reach $x=-1.80cm$ Time taken $= \frac{T}{4}$

$$\therefore t = \frac{T}{4}$$

$$= \frac{0.2}{4}$$

$$= \frac{1}{20} \text{ sec}$$

∴ Time taken to go from $x=0$ to $x=-1.8$ cm $t = \frac{1}{20}$ sec

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Problem: 14.16:

A 0.400kg Object undergoing SHM has $a_x = -2.70ms^{-2}$ when $x = 0.300m$. What is the time for the oscillation?

⇒ In given:

$m = 0.400kg$

$a_x = -2.70ms^{-2}$

$x = 0.300m$

We know:

$a_x = -\omega^2 x$

$-2.70 = -\omega^2 \times 0.300$

$\therefore \omega = \sqrt{\frac{2.70}{0.30}}$

$= 3 \text{ rad } s^{-1}$

Again: $T = \frac{2\pi}{\omega}$

$= \frac{2 \times \pi}{3}$

$= 2.0943 \text{ sec}$

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Problem: 14.17:

⇒ On a frictionless, horizontal air track. A glider oscillates at the end of an ideal spring of force constant 2.50 N/cm. The graph in Fig. E14.17 shows the accelerration of the glider as a function of time. Find.

(a) The mass of glider.

(b) The maximum displacement of the glider as a function from equilibrium point.

(c) The maximum force the spring exerts on the glider.

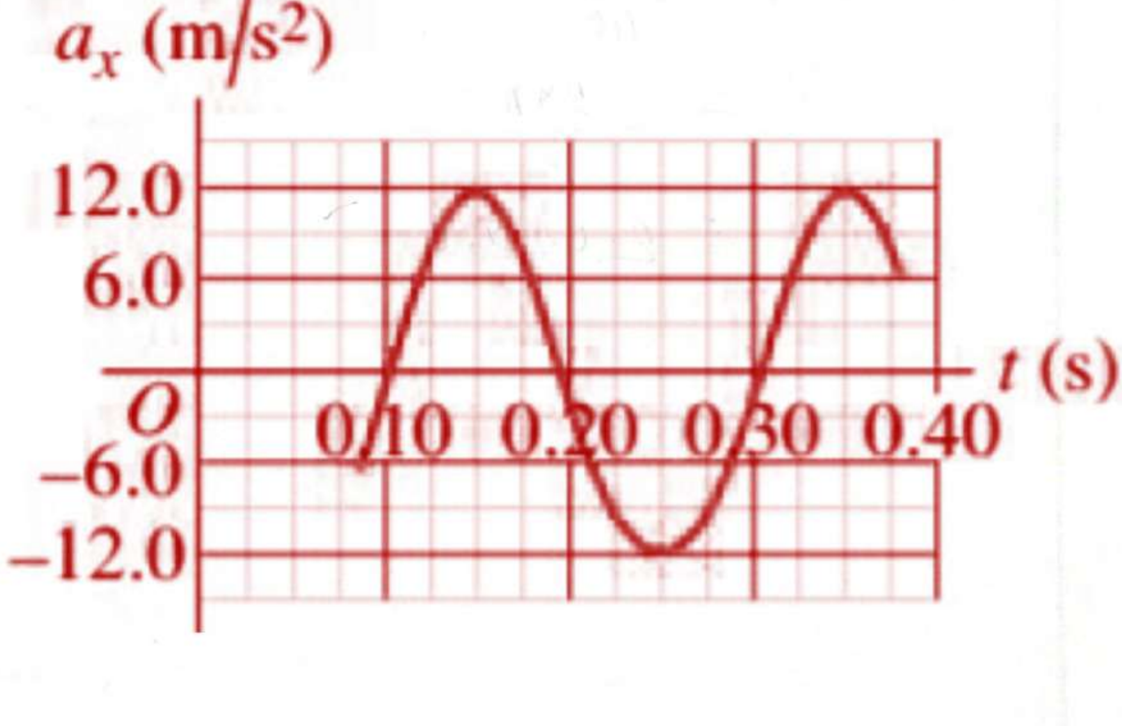

Figure **E14.17**

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⇒ In given:—

$$k = 2.50 N/cm = 2.50 \times 10^{2} N/m$$

$$= 250 N/m$$

$$a_{max} = 12 ms^{-2}$$

$$T = 0.20 sec$$

(a)

We know:—

$$T = \frac{2\pi}{\omega}$$

$$\therefore \omega = \frac{2\pi}{T}$$

$$= \frac{2\pi}{0.20}$$

$$= (10\pi) rad/s$$

Again:— $\omega^2 = \frac{k}{m}$

$$\therefore m = \frac{k}{\omega^2}$$

$$= \frac{250}{(10\pi)^2}$$

$$= 0.2533 kg$$

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(b)

We know:-

$$a_{max} = -\omega^2 x$$

$$\therefore x = -\frac{a_{max}}{\omega^2}$$

$$= \frac{-12.0}{(10\pi)^2}$$

$$= -0.012 \, m$$

$$= -1.21 \, cm$$

The minius sign means acceleration and Dinplecement Always have opposite sign.

Alternative:-

We know:-

$$F = -kx$$

$$ma = -kx$$

$$\therefore x = -\frac{ma}{k}$$

$$= -\frac{0.2533 \times 12}{250}$$

$$= -0.0121 \, m$$

$$= -1.21 \, cm$$

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(c)

We know:-

$$F = ma_{max}$$

$$= 0.2533 \times 12$$

$$= 3.0396 N$$

Alternative:-

We know:-

$$F = -kx$$

$$= -(250) \times (-0.0121)$$

$$= 3.03 N$$

Problem: 14.18:-

⇒ A 0.51kg mass on a spring has velocity as a function of time given by $v_x(t) = -(3.60 \text{ cm/s}) \sin[(4.71 \text{ rad/s})t - \frac{\pi}{2}]$. What are.

(a) The Period

(b) The amplitude.

(c) The maximum acceleration of the mass

(d) The Force constant of the spring.

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⇒ We know:—

The standard equation for Velocity of SHM:—

$$V(t) = -A\omega \sin(\omega t + \phi)$$

In given:—

$$V(t) = -(3.60\,cm/s)\sin[(4.71\,rad/s)t - \frac{\pi}{2}]$$

(a)

Comparing this two equation we get:—

$$\omega = 4.71\,rad/s$$

We know:—

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4.71}$$

$$= 1.33\,sec$$

(b)

Comparing this two equation, so,

$$A\omega = 3.60\,cm/s \quad \text{——— (i)}$$

$$\omega = 4.71\,rad/s \quad \text{——— (ii)}$$

Substitute the value of ω in equation (i) ⇒

$$A \times 4.17 = 3.60$$

$$\therefore A = \frac{3.60}{4.71}\,cm$$

$$= 0.76\,cm = 7.6 \times 10^{-3}\,m$$

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(C)

We have:—

$\omega = 4.71$ rad/s

$A = 7.6 \times 10^{-3}$ m

We know:—

$a_{max} = A\omega^2$

$= (7.6 \times 10^{-3}) \times (4.71)^2$

$= 0.16\ ms^{-2}$

$= 16.85\ cms^{-2}$

(D)

We know:—

$K = \omega^2 m$

$= (4.71)^2 \times 0.500$

$= 11.09\ N$

$m = 0.500\ kg$

$\omega = 4.71$ rad/s

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Problem: 14.19:-

⇒ A 1.50 kg mass on a spring has displacement as function of time given by the equation.

$$x(t) = (7.40\,cm)\cos[(4.16\,rad/s)t - 2.42]$$

Find:-

(a) The Time for one complete Vibration.

(b) The force constant of the spring.

(c) The maximum speed of the mass.

(d) The maximum force of the mass.

(e) The position, speed, and accelaration of the the mass at $t = 1.00s$

(f) The force on the mass at that time.

⇒ We know:-

The standard equation of SHM for Displacement:-

$$x(t) = A\cos(\omega t + \phi)$$

In given:

$$x(t) = (7.40\,cm)\cos[(4.16\,rad/s)t - 2.42]$$

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(a)

Comparing this two equation we get:—

$\omega = 4.16 \ rad/s$

$A = 7.40 cm = 7.40 \times 10^{-2} m$

We know:—

$$T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{4.16}$$

$$= 1.51 sec$$

(b)

We know:—

$$K = \omega^2 m = (4.16)^2 \times 1.50$$

$$= 25.95 \ N/m$$

$m = 1.50 kg$

$\omega = 4.16 \ rad/s$

(c)

We know:—

$$V_{max} = \omega A$$

$$= 4.16 \times 7.40 \times 10^{-2}$$

$$= 0.308 ms^{-1}$$

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(d)

We know:—

$$F_{max} = m a_{max} \quad \text{——— (1)}$$

$$a_{max} = \omega^2 A$$

$$= (4.16)^2 \times 7.4 \times 10^{-2}$$

$$= 1.28 \ ms^{-2}$$

Substitute the value of a_{max} in equation (1) ⇒

$$\therefore F_{max} = m a_m$$

$$= 1.50 \times 1.28$$

$$= 1.92 N$$

(e)

It's given:—

$t = 1 sec$

The position:—

$$x(t) = (7.40 cm) \cdot \cos[(\omega t + \phi)]$$

$$= (7.40 cm) \cos[(4.16) \times 1 - 2.42]$$

$$= -1.24 \ cm$$

The speed:— $V(t) = -A\omega \sin(\omega t + \phi)$

$$= -(7.40 \times 4.16) \sin[((4.16 \times 1) - 2.42)]$$

$$= -30.3 \ cm/s$$

$$= -0.303 ms^{-1}$$

the minius sign endicate the direction of SHM

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The acceleration:— $a(t) = -A\omega^2 \cos(\omega t + \phi)$

$= -(7.49) \times (4.16)^2 \cos[(4.16) \times 1 - 2.42]$

$= 21.56 \ cm/s^2$

$= 0.2156 \ ms^{-2}$

(F)

We have:—

$a = 0.2156 \ ms^{-2}$

$m = 1.50 \ kg$

We know:—

$F = ma$

$= 1.5 \times 0.2156$

$= 0.3234 \ N$

Ans:—

14.22:- Problem:—

⇒ A small block is attached to an ideal spring and is moving in SHM on a Horizontal frictionless surface. The Amplitude of the motion is 0.120m. The maximum speed of the block is 3.90 ms⁻¹. What the maximum acceleration of the block.

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⇒ In given:—

$A = 0.120 m$.

$V_{max} = 3.90 ms^{-1}$

We know:—

$V_{max} = A\omega$

$\therefore \omega = \frac{V_{max}}{A} = \frac{3.90}{0.120}$

$= 32.5 rad/s$

$a_{max} = A\omega^2$

$= (0.120)(32.5)^2$

$= 126.75 ms^{-2}$

Ans!

Problem 14.24:—

⇒ A small block is attached to an Ideal spring and is moving in SHM on a Horizontal, frictionless surface. The amplitude of the motion is 0.250 m and the period is 3.20 s.

(a) What are the speed and

(b) acceleration of the block when $x = 0.160 m$?

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⇒ Is given:—

$A = 0.250 m$

$T = 3.20 \sec$

$x = 0.160 m$

We know:— (a)

$T = \frac{2\pi}{\omega}$

$\therefore \omega = \frac{2\pi}{T} = \frac{2\pi}{3.20}$

$= 1.96 \text{ rad/s}$

$V = \omega\sqrt{A^2 - x^2}$

$= 1.96\sqrt{(0.250)^2 - (0.160)^2}$

$= 0.377 ms^{-1}$

(b)

We know:—

$a = -\omega^2 x$

$= -(1.96)^2 \times (0.160)$

$= -0.614 ms^{-2}$

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Problems 14.30:—

⇒ A 0.150kg toy is undergoing SHM on the end of a Horizontal spring with force constant k=300N/m. When the object is 0.0120m from its equilibrium position, it is observed to have a speed of $0.300ms^{-1}$ What are.

(a) The total Energy of the object at any point of its motion.

(b) The Amplitude of the motion.

(c) The maximum speed attained by the object during its motion.

⇒ In given:—

$m = 0.150 kg$

$K = 300 N/m$

$x = 0.0120 m$

$V = 0.300 ms^{-1}$

(a)

We know:—

$U = \frac{1}{2}kx^2$

$= \frac{1}{2} \times 300 \times (0.0120)^2$

$= 0.0216 J$

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$E_k = \frac{1}{2}mv^2$

$= \frac{1}{2} \times (0.150) \times (0.300)^2$

$= 6.75 \times 10^{-3}$ J

Total Energy $E = U + E_k$

$= 0.0216 + 6.75 \times 10^{-3}$

$= 0.02835$ J

(b)

We know,

$E = \frac{1}{2}kA^2$

$\therefore A = \sqrt{\frac{2E}{k}}$

$= \sqrt{\frac{2 \times 0.02835}{300}}$

$= 0.0137$ m

(c)

We know,

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{300}{0.150}} = 20\sqrt{5}$ rad/s

$V_{max} = \omega A$

$= 20\sqrt{5} \times 0.0137$

$= 0.6126\ ms^{-1}$

Ans

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Waves:—

⇒ Wave is a propagation of disturbances from place to place in a regular and organized way.

Some Example of wave:—

1. Ripples on a pond
2. musical sounds.
3. Light from the sun.

⇒ As a wave propagates it carries energy.

Electromagnetic waves:—

⇒ Electromagnetic waves that are created as a result of vibrations between an electric field and magnetic field.

Example:—

1. Light
2. Radio
3. X-ray etc.

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Mechanical Waves:—

⇒ Waves that travels within some medium called Mechanical waves.

Example:—

1. Sound waves.
2. Water waves.

Types of Mechanical wave:—

⇒ There are three types of Mechanical Wave (MW)

1. Transverse wave.
2. Longitudinal wave.
3. Surface wave.

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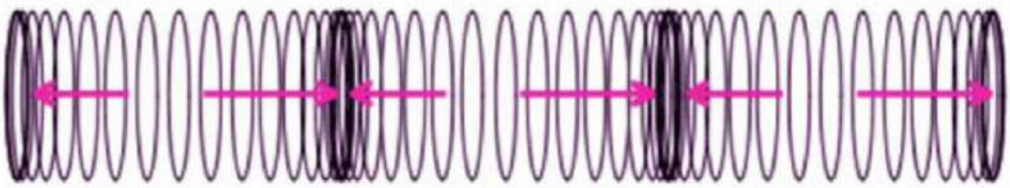
Transverse Wave:-

=> When the movement of the particles is at right Angeles or perpendicular to the motion of the energy, then this type of wave is known as a transverse wave. Ex:- Light.

Transverse Wave

Longitudinal Wave:-

=> When the movement of the particles is parallel to the motion, of the energy. and displacement of the medium is the same direction, in which the wave is moving is known as Longitudinal wave. Ex:- Sound.



Longitudinal Wave

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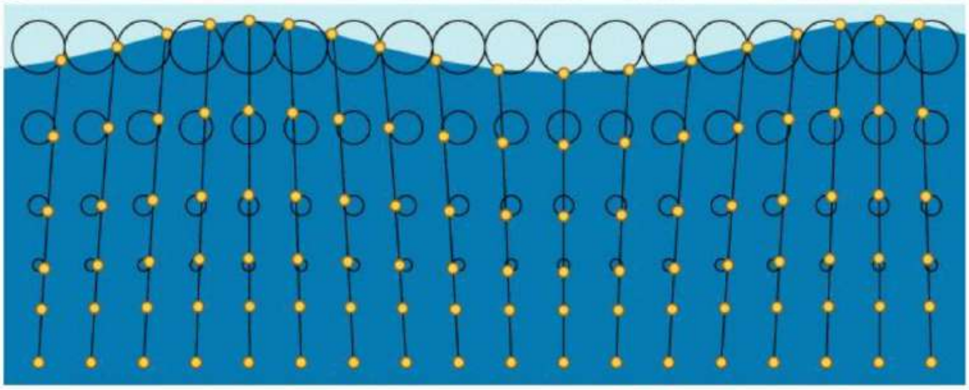
Surface Wave:-

=> In thin type wave, the particles travel in a Circular motion. These wave usally occur at interface.



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Properties of Mechanical Waves:-

(1) Wave speed:-

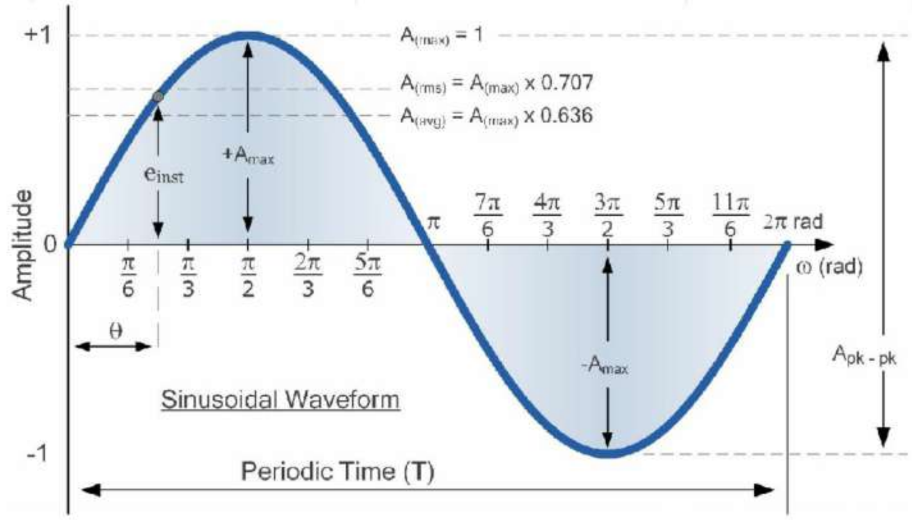
=> The disturbance travels with a definite speed called wave speed through the medium.

(2) The medium itself does not travel through space, It individual particle undergoes back-and-fort or up and down motions around there equilibrium position.

(3) The wave motion transport energy from one region of the medium to other.

#Sinusoidal Waves:-

⇒ Periodic waves with SHM are called Sinusoidal Waves.



1. The wave pattern travels with constant speed V and advances a dintance of one weivelength λ in a time interval of one Time period T. so, The wave speed is

$$\boxed{V = \frac{\lambda}{T} = \lambda f}$$

$$\boxed{f = \frac{1}{T}}$$

V = Wave speed
λ = wavelength
T = time period.
f = frequency.

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Problem: 15.1:-

⇒ Sound waves are longitudinal waves in air. The speed of sound depends on temperature at 20°c, it is $344 ms^{-1}$. What is the wavelength of a sound wave in air at 20°c if the frequency is 262 Hz.

⇒ Is given:-

$V = 344 ms^{-1}$

$f = 262 Hz$

We know:-

$V = \lambda f$

$\therefore \lambda = \frac{V}{f}$

$= \frac{344}{262}$

$= 1.31 m$

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Problem 15.21 —

⇒ If you double the wave length of a wave on a particular string. what happens to the wave speed v and the frequency f.

⇒ (iv) v is unchanged and f becomes one-half as great.

Mathematical descriptions of a Wave:—

A sinusoidal wave can be mathematically represented as:

(1) $y(x,t) = A\cos(kx - \omega t)$ → moves in $+x$ direction.

(2) $y(x,t) = A\cos(kx + \omega t)$ → moves in $-x$ direction.

where—

$y(x,t)$ = Displacement of the wave at position x and time t.

A = Amplitude.

$k = \frac{2\pi}{\lambda}$ = Wave number. rad/m.

x = position variable along the direction of wave propagation

$\omega = 2\pi f$ = Angular frequency.

t = Time

ϕ = Phase angle.

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(1) For a travelling sinusoidal waves —

$$y(x,t) = A\cos(kx \pm \omega t)$$

Where, $(kx \pm \omega t)$ = Phase. (measured in radians).

Speed of a Sinusoidal wave:—

⇒ The wave speed is the speed with which a phase of the waves moves. It also called phase speed.

⇒ For a particular phase the transverse displacement remains always constant and its possible only if the phase angle remains constant. Thus for a sinusoidal wave travelling in the $+x$ direction.

$$\therefore (kx - \omega t) = \text{constant}.$$

$$\frac{d}{dt}(kx - \omega t) = 0$$

$$\Rightarrow \frac{dx}{dt} k - \frac{dt}{dt}\cdot\omega = 0$$

$$\Rightarrow \frac{dx}{dt} \times k = \omega$$

$$\Rightarrow \frac{dx}{dt} = \frac{\omega}{k}$$

$$\therefore \boxed{V = \frac{\omega}{k}}$$

$$\boxed{V = -\frac{\omega}{k}}$$ → moves in $-x$ direction.

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$$\therefore V = \frac{2\pi f}{\frac{2\pi}{\lambda}}$$

$$\boxed{\therefore V = f\lambda}$$ → move in +x direction.

$$\boxed{V = -f\lambda}$$ → moves in +x direction.

problem 1001 (2)

⇒ A sinusoidal wave travels along a string. The time for a particular point to move from maximum displacement to zero in 0.170s. What are

(a) The time period.

(b) the frequency.

(c) The wavelength in 1.40m; what are the wave speed.

⇒ In given:— (a)

$t_1 = 0$ sec

$t_2 = 0.170$ sec

$$\Rightarrow \frac{T}{4} = (t_2 - t_1)$$

$$= (0.170 - 0)$$

$$\therefore T = 0.170 \times 4$$

$$= 0.68 \text{sec}.$$

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(b)

We know:—

$$f = \frac{1}{T}$$

$$= \frac{1}{0.68}$$

$$= 1.47 Hz$$

(c)

In given:—

$$\lambda = 1.40 m$$

We know:—

$$v = f\lambda$$

$$= 1.47 \times 1.40$$

$$= 2.436 ms^{-1}$$

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The Wave Equation:-

1. We know,

Transverse Displacement $y(x,t) = A\cos(kx - \omega t) \rightarrow$ along $+x$

2. Acceleration:-
$$\frac{\partial^2}{\partial t^2} y(x,t) = -\omega^2 A\cos(kx - \omega t)$$
$$= -\omega^2 y(x,t) \quad \text{—— (I)}$$

3. 2nd partial derivatives with respect to x:-
$$\frac{\partial^2}{\partial x^2} y(x,t) = -k^2 A\cos(kx - \omega t)$$
$$= -k^2 y(x,t) \quad \text{—— (II)}$$

4. We know:-
$$\boxed{V = \frac{\omega}{k}}$$

Now, (I) ÷ (II) $\Rightarrow$

$$\frac{\partial^2/\partial t^2 \{y(x,t)\}}{\frac{\partial^2 \{y(x,t)\}}{\partial x^2}} = \frac{\omega^2}{k^2} = V^2$$

$$\therefore \boxed{\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{V^2} \times \frac{\partial^2 y(x,t)}{\partial t^2}} \rightarrow \text{(Wave equation)}$$

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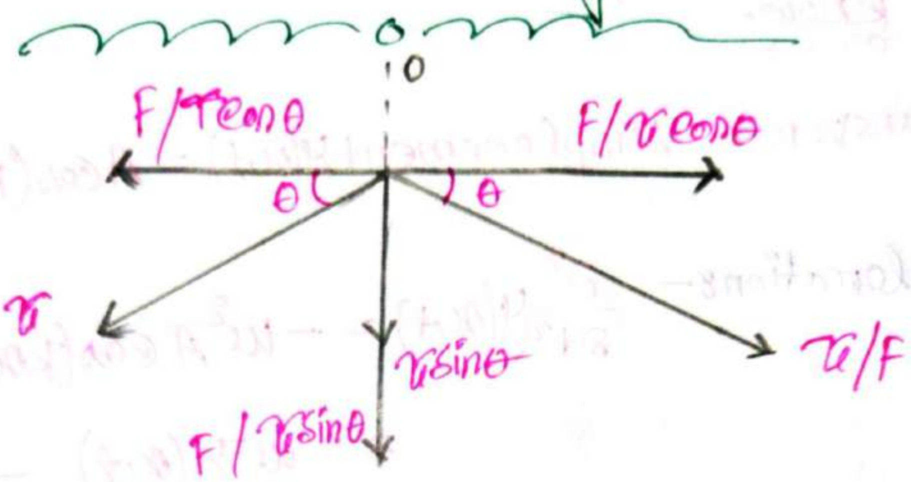
Wave speed on a Stretched String:



1. Let, A small string length = Δl

   mass = Δm

   Angle = 2θ

2. The Horizontal components ($\tau/F\cos\theta$ and $-\tau/F\cos\theta$) of τ cancels each other.

3. The resulting restoring Force.

$F' = 2\tau\sin\theta$

$= \tau 2\theta$

$$\boxed{F' = \tau\frac{\Delta l}{R}}$$

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4. The centripetal acceleration is:—

$$\boxed{a = \frac{v^2}{R}}$$

5. The mass of the element (Δl) is:—

$$\boxed{\Delta m = \mu \Delta l}$$

$$\boxed{\mu = \frac{\Delta m}{\Delta l}}$$

μ = mass per unit length.

6. Applying Newton's 2nd law of motion $(F = ma)$

$$F' = m\,a$$

$$\tau \frac{\Delta l}{R} = \mu \Delta l \times \frac{v^2}{R}$$

$$\boxed{\therefore\ v = \sqrt{\frac{\tau}{\mu}}}$$

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Problem 81:

⇒ The equation of transverse wave on a string is $y = (2 m)\cos[(20 m^{-1})x - (600 rad/s^{-1})t]$ The tension in the string $T = 15 N$.

(a) What is the wave speed.

(b) Find the linear density of this string in grams per meter.

⇒ is given:

$$y(x,t) = (2m)\cos[(20 m^{-1})x - (600 rad/s^{-1})t] \quad \text{---} (1)$$

$$T = 15 N$$

We know:

$$y(x,t) = A\cos(kx - \omega t) \quad \text{---} (11)$$

Now Compare equation (1) and (11) ⇒

$A = 2 m$

$k = 20 m^{-1}$

$\omega = 600 rad/s$

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(a)

We know:-

$$V = \frac{\omega}{k}$$

$$= \frac{600}{20}$$

$$= 30 ms^{-1}$$

(b)

We know:-

$$V = \sqrt{\frac{T}{\mu}}$$

$$\mu = \frac{T}{V^2}$$

$$\therefore \mu = \frac{15}{(30)^2}$$

$$= 0.0167 \, kg/m$$

$$= 16.7 \, gm/m$$

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Problem § 15.3:—

⇒ One end of a 2kg rope is tied to a support at the top of a mine shaft 80.0m deep. The The rope is stretched tault by a 20kg box of rocks attached at the bottom.

(a) The geologist at the bottom of the shaft signals to a colleague at the top by jerking the rope side ways. What the speed of a transverse wave on the rope?

(b) If a point on the rope is in transverse SHM with $f = 2Hz$. How many cycles of the wave are there in the rope's length?

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⇒ In given:-

$m_{rope} = 2\,kg$

$m_{box} = 20\,kg$

$\Delta l_{rope} = 80\,m$

We know:- (a)

Tension $T = m_{box} g$

$= 20 \times 9.8$

$= 196\,N$

$\mu = \frac{\Delta m_{rope}}{\Delta l_{rope}}$

$= \frac{2}{80}$

$= 0.025\,kg/m$

$\therefore V = \sqrt{\frac{T}{\mu}}$

$= \sqrt{\frac{196}{0.025}}$

$= 88.54\,ms^{-1}$

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(b)

We know:-

$$V = f\lambda$$

$$\therefore \lambda = \frac{V}{f}$$

$$= \frac{88.54}{2}$$

$$= 44.27m$$

λ = length of one wave

$$\therefore \text{Total number of cycle} = \frac{\text{Total length of roop}}{\text{length of one wave}}$$

$$= \frac{80m}{44.27}$$

$$= 1.8070m$$

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Problem : 15.4 :-

⇒ The six string of a guitar are the same length and under nearly the same tension. but they have different thickness. On which string do waves travel the fastest.

(i) The thickest string.

(ii) The thinnest string

(iii) The wave speed is the same on all strings.

⇒ We know:-

$$V = \sqrt{\frac{\tau}{\mu}} \qquad \tau = \text{constant}$$

$$V \propto \sqrt{\frac{1}{\mu}} \quad \text{——— (i)}$$

and we also know:-

$$\mu = \frac{\Delta m}{\Delta l} \qquad \Delta l = \text{constant}$$

$$\mu \propto \Delta m \quad \text{——— (ii)}$$

That means:-

$$V \propto \sqrt{\frac{1}{\Delta m}}$$

So, The thinnest string will have the largest speed of wave propagation.

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Wave Interference, boundary Condition and Superposition:-

⇒ When a propagates of wave through a medium, it reflects when it encounters the boundary of the medium. The wave before hitting the boundary in known as the incident wave. The wave After encountering the boundary in known as the reflected wave.

when a wave encounters a boundary, it can reflect back. the type of boundary condition (fixed or free) determines how the wave in reflected.

(1) Fixed Boundary: The wave reflects with an inverted phase.

(2) Free Boundary:- The wave reflects with the same phase.

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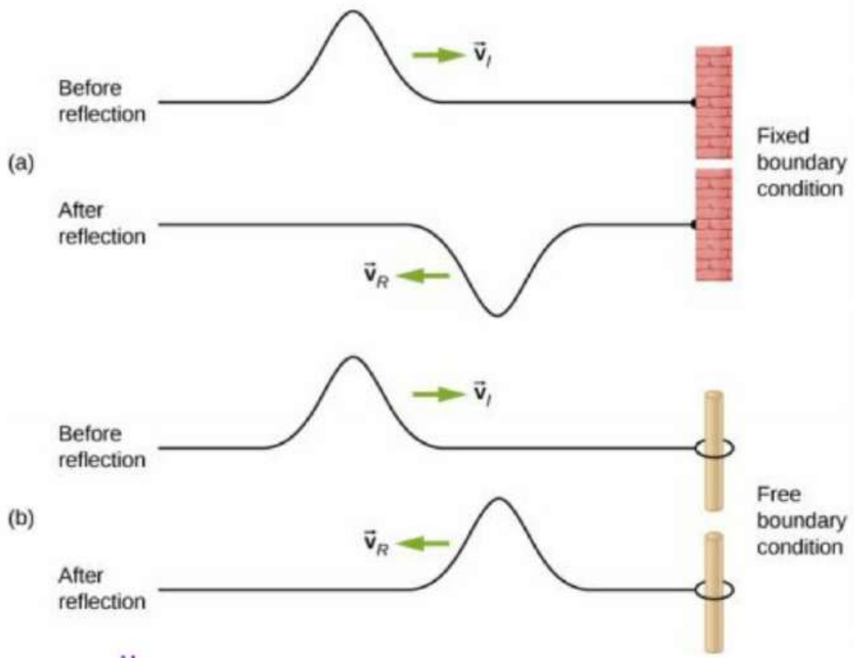

Interferencee:-

⇒ The initial wave and reflected waves overlaps in the same region of the medium. The overlapping of the waves is called the interference.

Superposition principle:-

⇒ The transverse displacement of the medium is the algebric sum of the transverse displacement of the individual waves.

∴

$$y(x,t) = y_1(x,t) + y_2(x,t)$$

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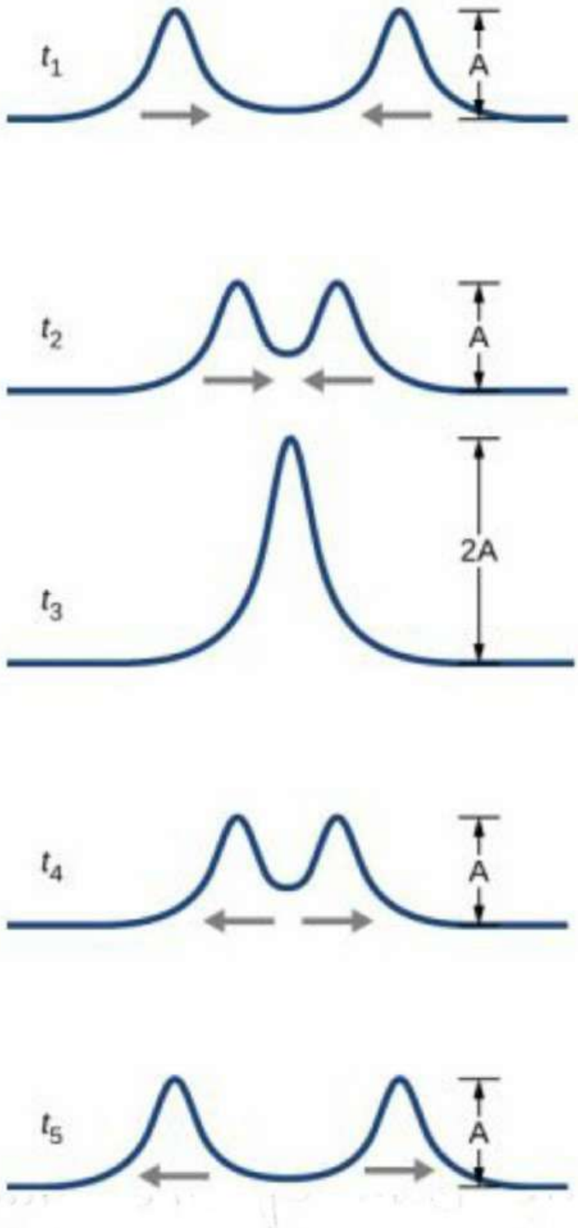
⇒ The principle of superposition can be used to analyze the combination of waves.



Interference of waves:—

⇒ Suppose we send two sinusoidal waves of the Same wavelength and Amplitude in the same direction along a stretched string.

$$y_1(x,t) = A \cos(kx - \omega t)$$

Now, Consider Another same wave is travelling with a different phase Angle.

$$\therefore y_2(x,t) = A \cos(kx - \omega t + \phi)$$

We know:—

Superposition principle:

$$y(x,t) = y_1(x,t) + y_2(x,t)$$

$$= A\cos(kx-\omega t) + A\cos(kx - \omega t + \phi)$$

$$= A\{\cos(kx-\omega t) + \cos(kx - \omega t + \phi)\}$$

$$= A\{2\cos\left(\frac{kx-\omega t+kx-\omega t+\phi}{2}\right)\cdot\cos\left(\frac{kx-\omega t-kx+\omega t-\phi}{2}\right)$$

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$$= 2A\left\{\cos\left(\frac{2(kx-\omega t)+\phi}{2}\right)\cdot\cos\left(\frac{-\phi}{2}\right)\right\}$$

$$= 2A\cos\left\{(kx-\omega t)+\frac{\phi}{2}\right\}\cdot\cos\left(\frac{-\phi}{2}\right)$$

$$y'(x,t) = \left[2A\cos\left(\frac{\phi}{2}\right)\right]\cos\left(kx-\omega t+\frac{\phi}{2}\right) \rightarrow \text{[For traveling waves]}$$

here:-

$y'(x,t)$ = Resultant Displacement.

$\left[2A\cos\left(\frac{\phi}{2}\right)\right]$ = Amplitude.

$\cos\left(kx-\omega t+\frac{\phi}{2}\right)$ = Oscillating term.

① if $\phi = 0$ rad $(0°)$: fully constructive interference.

$$y(x,t) = \left[2A\cos\left(\frac{0}{2}\right)\right]\cos\left(kx-\omega t+\frac{0}{2}\right)$$

$$= 2A\cos(0)\cdot\cos(kx-\omega t)$$

$$\therefore \boxed{y(x,t) = 2A\cos(kx-\omega t)} \rightarrow \text{[Highest Amplitude]}$$

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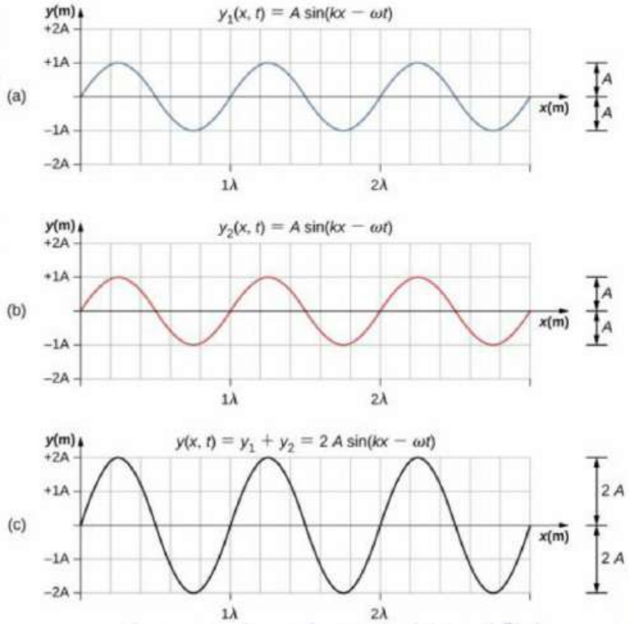


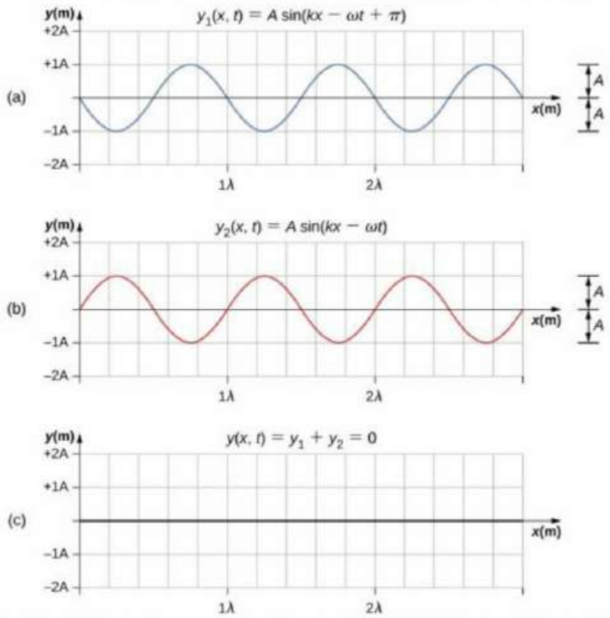
Constructive interference of two identical waves produces a wave with twice the amplitude, but the same wavelength.

(2) If $\phi = \pi$ rad (180°) fully destructive interference.

$$y(x,t) = \left[2A\cos\left(\frac{\pi}{2}\right)\right]\cos\left(kx - \omega t + \frac{\pi}{2}\right)$$

$$= \left[2A \times 0\right] \times \cos\left(kx - \omega t + \frac{\pi}{2}\right)$$

$$\boxed{y(x,t) = 0}$$



Destructive interference of two identical waves, one with a phase shift of 180°(π rad), produces zero amplitude, or complete cancellation.

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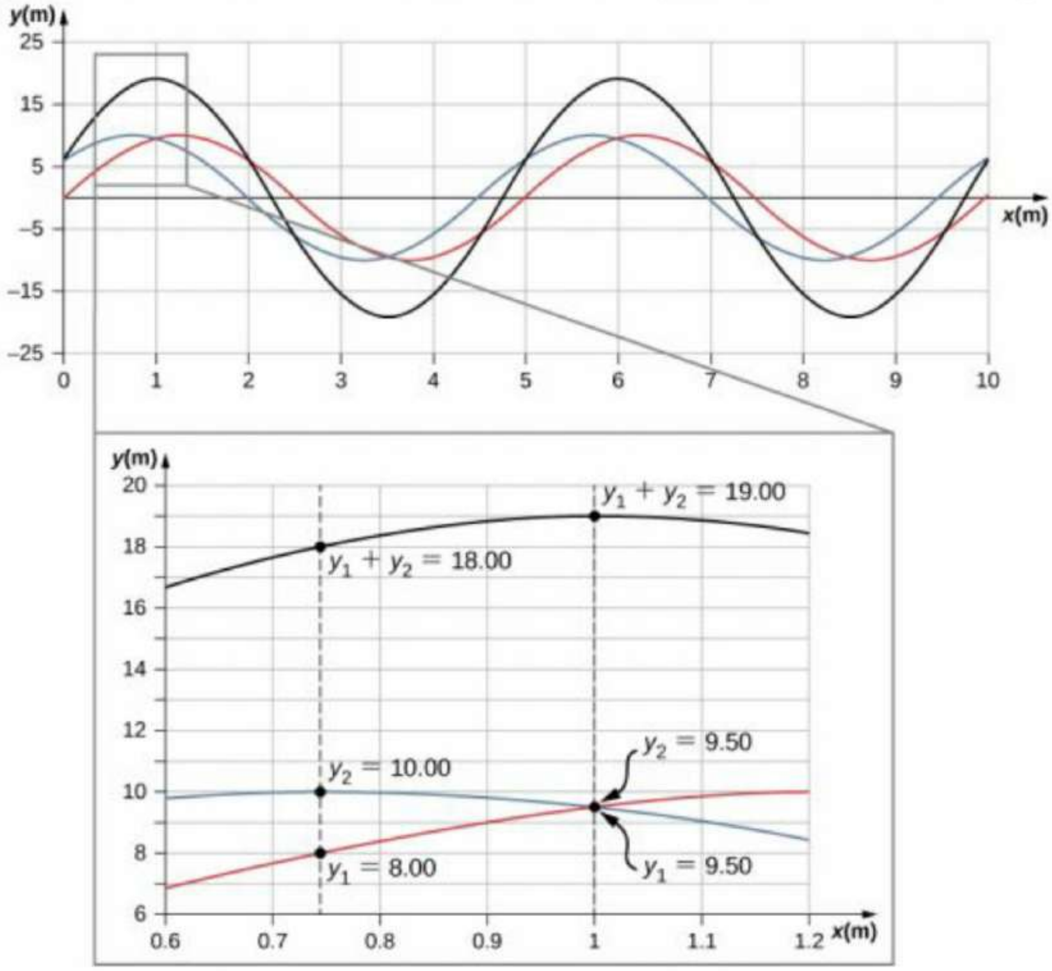
3. If $\phi = \frac{2\pi}{3}$ rad (120°):- Intermediate Interference:-

$$y(x,t) = \left[2A\cos\left(\frac{2\pi/3}{2}\right)\right]\cos\left(kx-\omega t+\frac{2\pi/3}{2}\right)$$

$$= 2A\cos\left(\frac{\pi}{3}\right)\cos\left(kx-\omega t+\frac{\pi}{3}\right)$$

$$= 2A \times \frac{1}{2}\cos\left(kx-\omega t+\frac{\pi}{3}\right)$$

$$\boxed{y(x,t) = A\cos\left(kx-\omega t+\frac{\pi}{3}\right)}$$



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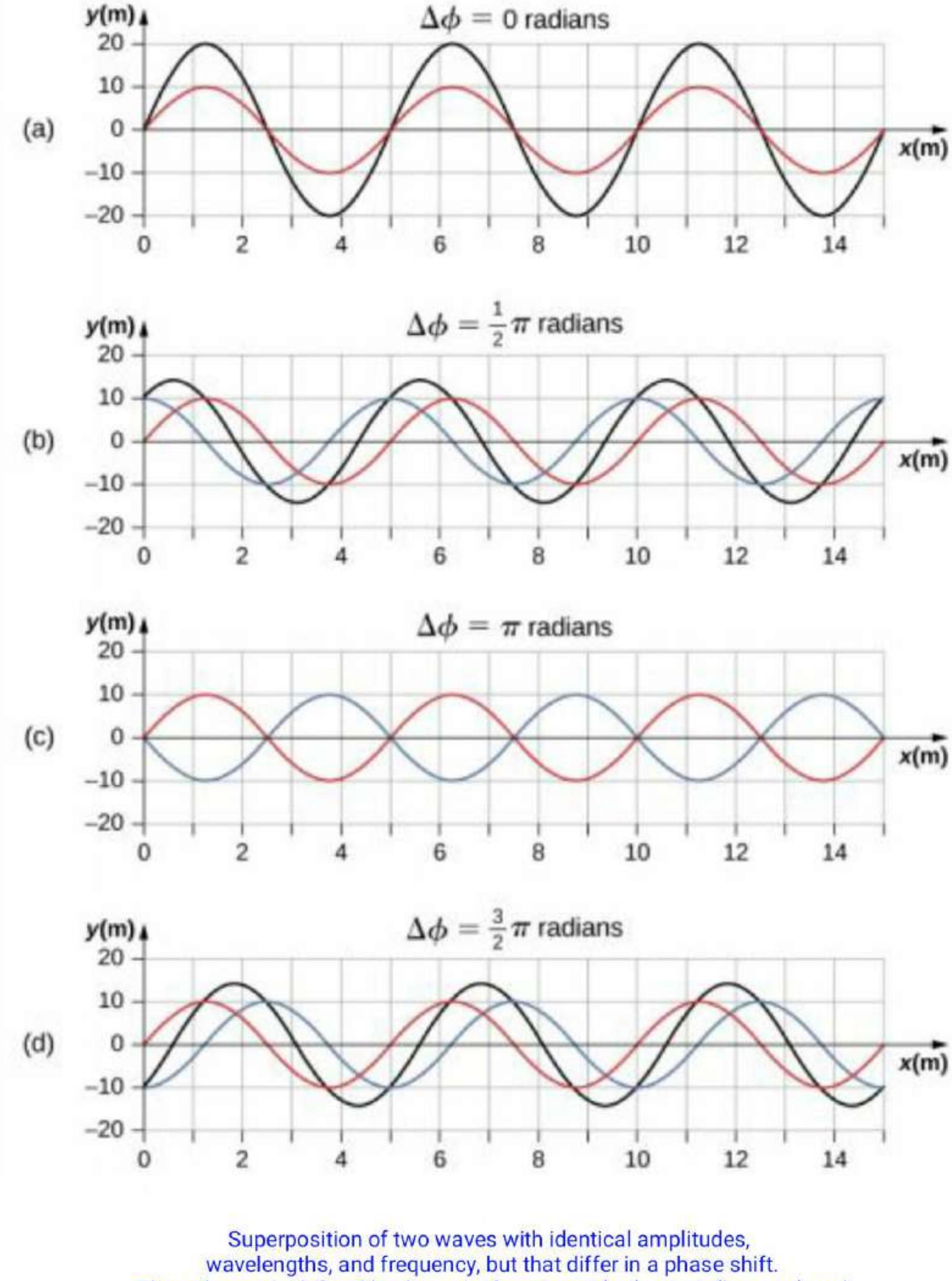
Some wave Interference patterns:-



Superposition of two waves with identical amplitudes, wavelengths, and frequency, but that differ in a phase shift. The red wave is defined by the wave function $y1(x, t) = A \sin(kx - \omega t)$ and the blue wave is defined by the wave function $y2(x, t) = A \sin(kx - \omega t + \phi)$. The black line shows the result of adding the two waves. The phase difference between the two waves are (a) 0.00 rad, (b) $\pi/2$ rad, (c) π rad, and (d) $3\pi/2$rad.

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Question:-

⇒ What phase difference between two identical traveling waves. moving in the same direction along a Stretched string. Results in the combined wave having an amplitude 1.50 times that of the common amplitude of the two combining waves?. Express the Answer in

(a) Degrees.

(b) radians.

(c) wavelengths.

⇒ we know:-

$$y(x,t) = \left[2A\cos\frac{\phi}{2}\right]\cos\left(kx - \omega t + \frac{\phi}{2}\right)$$

is given:- (a)

Amplitude of resulting wave = 1.50A

So,

$$2A\cos\frac{\phi}{2} = 1.50A$$

$$\phi = 2\cos^{-1}\left(\frac{1.50}{2}\right)$$

$$= 82.81^{\circ}$$

(b)

Converting radians:-

$$\phi = 82.81^{\circ} \times \left(\frac{\pi}{180}\right) \text{rad}$$

$$= 1.445 \text{ rad}.$$

(3)

We know:—

$$2\pi \text{ rad} = \lambda$$

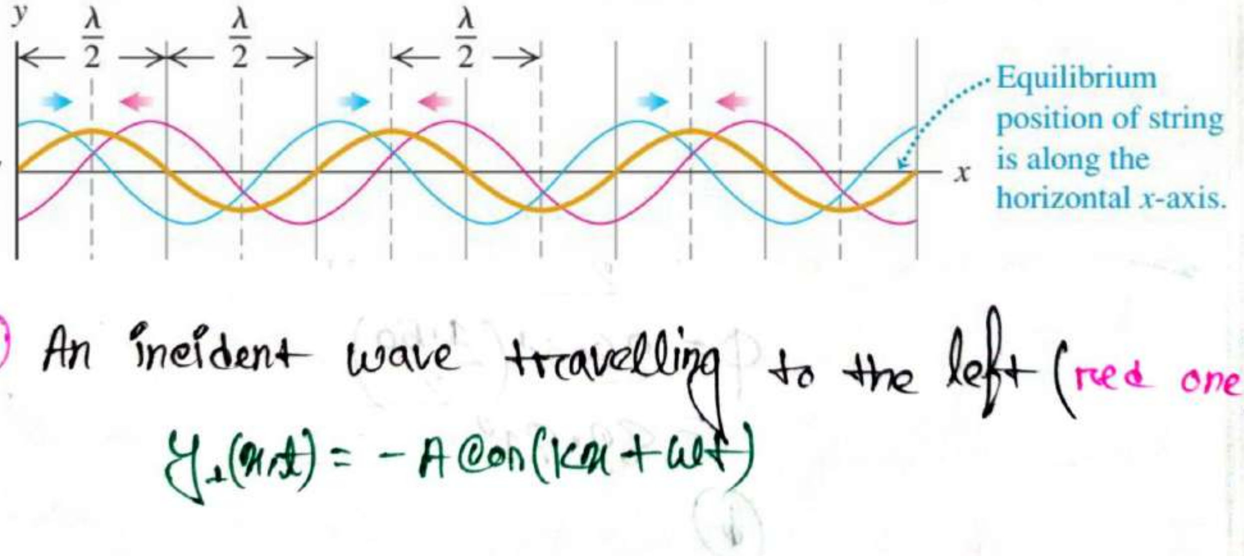
$$\therefore \lambda = 2\times\pi \text{ rad}$$

$$\therefore 1 \text{ rad} = \left(\frac{\lambda}{2\pi}\right)$$

In terms of wavelength (the length of cycle, where cycle corresponds to 2π rad)

this is equivalent to. $\frac{1.45}{2\pi} = 0.230$ wavelength

Standing wave along a string:—



(1) An incident wave travelling to the left (red one):—

$$y_1(x,t) = -A\cos(kx+\omega t)$$

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(2) Reflected wave travelling to the right (blue one):-

$$y_2(x,t) = A\cos(kx - \omega t)$$

(3) Applying Superposition principle:-

$$y(x,t) = y_1(x,t) + y_2(x,t)$$

$$= -A\cos(kx+\omega t) + A\cos(kx-\omega t)$$

$$= A[-\cos(kx+\omega t) + \cos(kx-\omega t)$$

$$= 2A\sin(kx)\cdot\sin(\omega t)$$

$$= (2A\sin kx)\sin(\omega t)$$

$$\boxed{\therefore y(x,t) = (2A\sin kx)\sin(\omega t)}$$

(4) So, The equation of Standing wave (Brown One):-

$$y(x,t) = (2A\sin kx)\cdot\sin(\omega t)$$

(5) Amplitude at position x: $2A\sin kx$

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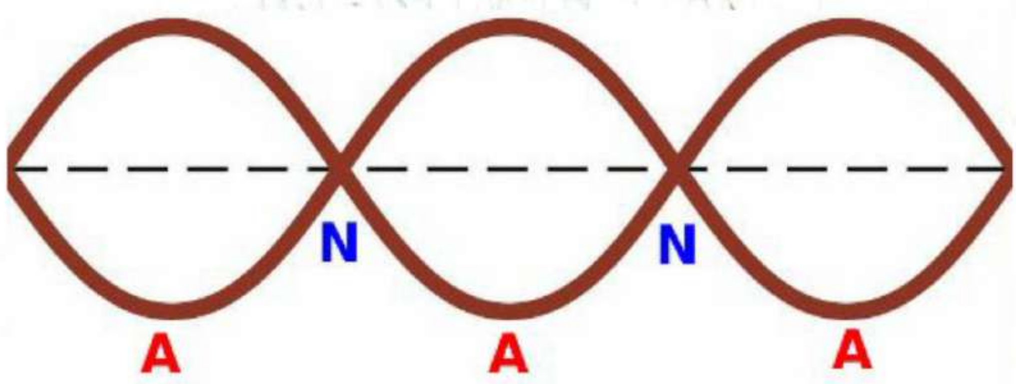
Standing Waves on a String:-



A=Antinode, N=Node

(1.) Nodes:-

⇒ Some points of the string never oscillates, fully destructive interference occurs. That is called Nodes denote by N

Antinodes:-

⇒ Some points oscillates with amplitudes 2A. fully constructive interference occurs, That is called Antinodes denote by AN

(2.) Each point in the string still undergoes SHM but all the point between any successive pair of nodes oscillate in phase difference between adjacent particles.

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(3) Standing wave, unlike a traveling wave, does not transfer energy from one to end to the other.

positions of Nodes and Antinodes:-

We know:-

The equation of Standing wave:-

$$y(x,t) = \underline{2A\sin(kx)} \cdot \sin(\omega t)$$

For Nodes:-

$\Rightarrow$ for any time t, $y(x,t) = 0$ it's possible only when

$$\sin(kx) = 0$$

This occur only when $kx = 0, \pi, 2\pi, 3\pi, \dots, n\pi$

and we know:-

$$k = \frac{2\pi}{\lambda}$$

$\therefore$ The position of Nodes:

$$x = 0, \frac{\pi}{\frac{2\pi}{\lambda}}, \frac{2\pi}{\frac{2\pi}{\lambda}}, \frac{3\pi}{\frac{2\pi}{\lambda}}, \dots, \frac{n\pi}{\frac{2\pi}{\lambda}}$$

$$\therefore \boxed{x = 0, \frac{\lambda}{2}, \frac{2\lambda}{2}, \frac{3\lambda}{2}, \dots \dots, \frac{n\lambda}{2}}, \boxed{n = 0, 1, 2, 3}$$

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For Antinodes:—

⇒ Between every two nodes, at the midway, we get one antinode.

So, The position of the Antinodes:—

$$\boxed{x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots, \left(n+\frac{1}{2}\right)\frac{\lambda}{2}}, \quad \boxed{n = 0, 1, 2, 3, \ldots}$$

problem 01:—

⇒ A String Oscillate according to the equation

$$y = (0.50\,\text{cm}) \sin\left[\left(\frac{\pi}{3}\,\text{cm}^{-1}\right)x\right] \sin\left[(40\pi\,\text{rad s}^{-1})t\right]$$

What are,

(a) The Amplitude.

(b) The speed of the two waves (identical except for direction of travel) whose superposition gives this oscillation.

(c) What is the distance between nodes?

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⇒ In given:-

$$y = (0.50\,cm)\,\sin[(\frac{\pi}{3}\,cm^{-1})x]\,\sin[(40\pi\,rads^{-1})t]$$

We know:-

$y = (2A\sin kx)\sin\omega t$ Compare both of them we get.

So,

$$K = \frac{\pi}{3}\ rad/cm$$

$$= \frac{\pi}{3} \times 100\ rad/m$$

$$K = 33.33\pi\ rad/m$$

$$\omega = 40\pi\ rad/s$$

(a)

$$\therefore\ 2A = 0.50cm$$

$$\therefore A = \frac{0.5\times10^{-2}}{2}$$

$$= 2.5\times10^{-3}m$$

(b)

We know:-

$$V = \frac{\omega}{K}$$

$$= \frac{40\pi}{33.33\pi} = 1.20\,ms^{-1}$$

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(C)

We know:-

$$\Delta x = \frac{\lambda}{2}$$

$$= \frac{\frac{2\pi}{K}}{2}$$

$$= \frac{\pi}{K}$$

$$= \frac{\pi}{33.33\pi}$$

$$= 0.03 m$$

$$K = \frac{2\pi}{\lambda}$$

$$\therefore \lambda = \frac{2\pi}{K}$$

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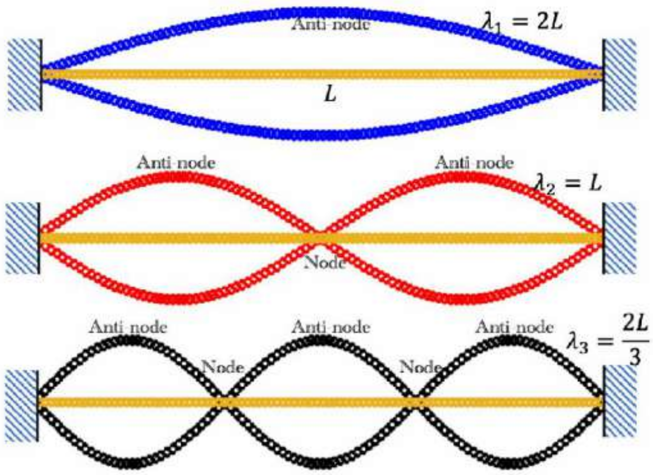
Normal Modes of a String:—



(1.) If a string with length L is fixed in both ends, A Standing wave can exist only if its wavelengths satisfies.

$$\boxed{\therefore L = n\frac{\lambda}{2},} \quad (n = 1, 2, 3, \ldots)$$

(2.) That's mean the Stable wavelengths:—

$$\boxed{\lambda_n = \frac{2L}{n}} \quad (n = 1, 2, 3, \ldots)$$

(3.) ∴ The Stable frequencies:—

$$\boxed{f_n = \frac{V}{\lambda_n} = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$$

f_n = harmonics/ resonant wavelengths/ frequencies.

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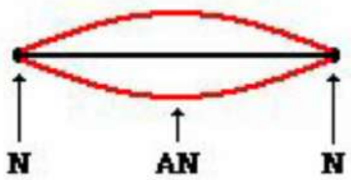
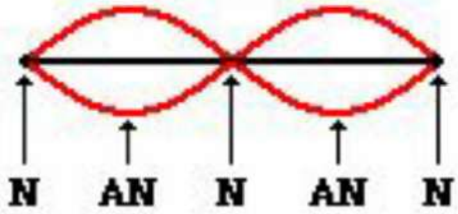
Harmonic Series:—

⇒ The sequence of harmonics is called harmonic series.

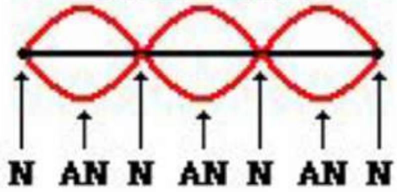
Fundamental Frequency or 1st Harmonic



2nd Harmonic



3rd Harmonic



Figures shows the first 4 normal modes of a string fixed at both ends.

(1.) Fundamental modes:—

⇒ Where, $n = 1$, nodes $= 2$, Antinodes $= 1$

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(2.) 2nd Harmonies:-

where, $n=2$, nodes $=3$, Antinodes $=2$

(3.) third Harmonie:-

Where, $n=3$. nodes $=4$, Antinodes $=3$

So,

$$\boxed{\lambda_n = \frac{2L}{n}}$$

$$\therefore \boxed{f_n = \frac{v}{\lambda_n} = \frac{n}{2L}\sqrt{\frac{\tau}{\mu}}}$$

Problem-1:-

=> 100cm length of A string in stretched between fixed supports. What are:

(a) longest

(b) 2nd longest

(c) 3rd longest wavelength for waves traveling on the string if standing waves are to be setup.

(d) Sketch those standing waves.

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⇒ Is given:—

$$L = 120\ cm = 120 \times 10^{-2} m$$

(a)

$n = 1$ for longest wavelength.

We know:—

$$\lambda = \frac{2L}{n}$$

$$= \frac{2 \times 120 \times 10^{-2}}{1}$$

$$= 2.4 m$$

Ans:—

(b)

$n = 2$

We know:—

$$\lambda = \frac{2L}{n}$$

$$= \frac{2 \times 120 \times 10^{-2}}{2}$$

$$= 1.2 m$$

Ans:—

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(c)

$n = 3$

We know:—

$$\lambda_n = \frac{2L}{n}$$

$$= \frac{2 \times 120 \times 10^{-2}}{3}$$

$$= 0.80\,m$$

Ans:—

(d)

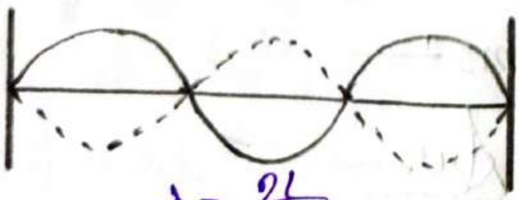
First Harmonic:—

$$\lambda = \frac{2L}{1}$$

Second Harmonic:—

$$\lambda = \frac{2L}{2}$$

Third Harmonic:—



$$\lambda = \frac{2L}{3}$$

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Problem! 2:-

⇒ A 125 cm length of a string has a mass 2 gram and tension 7 N between fixed supports.

(a) What is the wave speed for this string?

(b) What is the lowest resonant frequency of this string.

⇒ In given:-

$$L = 125 \times 10^{-2} m$$
$$\tau = 7 N$$
$$m = 2 \times 10^{-3} kg$$

(a)

We know:-

$$\mu = \frac{m}{L} = \frac{2 \times 10^{-3}}{125 \times 10^{-2}}$$
$$= 1.6 \times 10^{-3} kg/m$$

$$v = \sqrt{\frac{\tau}{\mu}}$$
$$= \sqrt{\frac{7}{1.6 \times 10^{-3}}}$$
$$= 66.14 ms^{-1}$$

Ans

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(b)

For the lowest resonant frequency $n=1$

We know:—

$$f = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$$

$$= \frac{1}{2L}\sqrt{\frac{7}{1.6\times10^{-3}}}$$

$$= \cancel{10\sqrt{7}\ Hz}$$

$$= 26.45 Hz$$

Ans:—

Problem: 15.3:—

⇒ On December 26, 2004, a great earthquake occurred off the coast of Sumatra and triggered immense wave that killed some 200,000 people. Satellites observing this waves from space measured 800 km from one wave crest to the next and a period between waves of 1.0 hour. What was the

(a) speed of these waves in ms^{-1} and kmh^{-1}

(b) Does your Ans help you understand why the waves caused such devastation?

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⇒ In given:-

$\lambda = 800 km = 800 \times 10^3 m$

$T = 1h = 3600 sec$

∴ We know:- (a)

$V = \frac{\lambda}{T}$

$= \frac{800}{1}$

$\boxed{V = 800 kmh^{-1}}$

$\therefore V = \frac{800 \times 10^3}{3600}$

$\boxed{V = 222.22 ms^{-1}}$

Ans:-

(b)

⇒ The tsunami's rapid speed led to powerful waves that struck coasts with tremendous force, causing immense devastation.

⇒ The High speed contributed to the devastating impact.

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Problem: 15.4:-

⇒ Sound having frequencies above the range of human hearing (20,000 Hz) is called ultrasound. Wave above this frequency can be used to penetrate the body and produce images by reflecting from surfaces. In a typical ultrasound scan 1500 ms^{-1}. For a good, detailed image. The wavelength should be no more than 1.0 mm.

(a) What frequency sound is required for a good scan?

⇒ To given:-

$V = 1500\ ms^{-1}$

$\lambda = 1mm = 1 \times 10^{-3} m$

We know-

$V = f\lambda$

$\therefore f = \frac{V}{\lambda}$

$= \frac{1500}{1 \times 10^{-3}}$

$= 1.5 \times 10^{6}\ Hz$

∴ For a good scan required frequency sound is 1.5×10^{6} Hz.

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Problem: 15.10:-

⇒ A water wave traveling in a straight line on a lake is described by the equation.

$$y(x,t) = (3.75\,cm)\cos(0.450\,cm^{-1}x + 5.40\,s^{-1}t)$$

where y is the displacement perpendicular to the disturbed surface of the lake.

(a) How much time does it take for one complete pattern to go past a fisherman in a boat at anchor, and what horizontal distance does the wave crest travel in that time.?

(b) What are the wave number and the number of wave per second that pass the fisherman.?

(c) How fast does a waves crest travel past the fisherman, what is the maximum speed of his cork floater as the wave causes it to bob up and down?

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⇒ In given:-

$$y(x,t) = (3.75\,cm)\cos(0.45\,cm^{-1}x + 5.40\,s^{-1}t)$$

$$y(x,t) = A\cos(kx + \omega t) \longrightarrow \text{Standard equation}$$

Comparing the two equation we get:-

$A = 3.75\,cm = 3.75 \times 10^{-2}\,m$

$k = 0.45\,cm^{-1} = 0.45 \times 100 = 45\,m^{-1}$

$\omega = 5.40\,rad\,s^{-1}$.

(i) (a)

We know:-

$$\omega = \frac{2\pi}{T}$$

$$\therefore T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{5.40}$$

$$= 1.163\ Sec$$ Ans.

(ii) We know:-

$$k = \frac{2\pi}{\lambda}$$

$$\therefore \lambda = \frac{2\pi}{k}$$

$$= \frac{2\pi}{45}$$

$$= 0.13\,m$$

$$= 13.96\,cm$$ Ans:-

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(b)

(i) We know:—

Wave number $K = 45$ rad m^{-1} / 0.45 rad/cm

Ans:—

(ii) We know:—

$\omega = 2\pi f$

$f = \frac{\omega}{2\pi}$

$= \frac{5.40}{2\times\pi}$

$= 0.859$ Hz / Ans:—

Alternative:—

We know:—

$f = \frac{1}{T}$

$= \frac{1}{1.163}$

$= 0.859$ Hz

Ans:—

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(c)

(i) We know,

$$V = \frac{\omega}{K}$$

$$= \frac{5.40}{45}$$

$$= 0.12\ ms^{-1}$$

$$= 12\ cms^{-1}$$

Ans:

Alternative

We know,

$$V = \frac{\lambda}{T}$$

$$= \frac{.13}{1.163}$$

$$= 0.12\ ms^{-1}$$

$$= 12\ cms^{-1}$$

Ans:

(ii) We know,

$$V_{max} = A\omega$$

$$= 3.75 \times 10^{-2} \times 5.40$$

$$= 0.2025\ ms^{-1}$$

$$= 20.25\ cms^{-1}$$

Ans:

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problem: 15.15:

⇒ One end of a horizontal rope is attached to a prong of an electrically driven tuning fork that vibrates the roop transversely at 120 Hz. The other end passes over a pulley and supports a 1.5 kg mass. The linear mass density of the rope is 0.0550 kg/m.

(a) What is the speed of transverse wave on the rope?

(b) What is the wave length?

(c) How would your answer to question (a) and (b) change if the mass were increased at 3 kg?

⇒ In given:

$f = 120\ Hz$

$m = 1.5\ kg$

$\mu = 0.0550\ kgm^{-1}$

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(a)

we know:-

$$T = mg$$

$$= 1.5 \times 9.8$$

$$= 14.7N$$

$$V = \sqrt{\frac{T}{\mu}}$$

$$= \sqrt{\frac{14.7}{0.0550}}$$

$$= 16.34\ ms^{-1}$$

Ans

(b)

we know:-

$$V = f\lambda$$

$$\therefore \lambda = \frac{V}{f}$$

$$= \frac{16.34}{120}$$

$$= 0.1361\ m$$

$$= 13.6167\ cm$$

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(C)

In given:—

$m' = 3\,kg$

(i) we know:—

$T' = m'g = 3 \times 9.8 = 29.4\,N$

$V' = \sqrt{\frac{T'}{\mu}} = \sqrt{\frac{29.4}{0.0550}} = 23.12\,ms^{-1}$

$\therefore \frac{V'}{V} = \frac{23.12}{16.34}$

$\therefore V' = 1.41V$

$\boxed{V' = \sqrt{2}V}$

(ii) we know:—

$\lambda' = \frac{V'}{f} = \frac{23.12}{120}$

$= 0.192\,m$

$\therefore \frac{\lambda'}{\lambda} = \frac{0.192}{0.1361}$

$\therefore \lambda' = 1.41\lambda$

$\boxed{\lambda' = \sqrt{2}\lambda}$

$\therefore$ if mass in doubled both speed and wavelength increases by a factor of $\sqrt{2}$.

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problem: 15.46:

⇒ With what tension must a rope with length 2.5m and mass 0.120 kg be stretched for transverse wave of frequency 40Hz to have a wavelength of 0.750m?

⇒ In given:

$L = 2.5m$

$m = 0.120kg$

$f = 40 Hz$

$\lambda = 0.750m$

We know:

$$v = \sqrt{\frac{\tau}{\mu}}$$

$$\therefore \tau = \mu v^2 \text{ ——— } (1)$$

$$\mu = \frac{m}{L} = \frac{0.120}{2.5} = 0.048 \, kgm^{-1}$$

$$v = f\lambda = 40 \times 0.750 = 30 \, ms^{-1}$$

Substitute the value of 'v' and 'μ' in eq (1) ⇒

$$\therefore \tau = \mu v^2$$
$$= 0.048 \times (30)^2$$
$$= 43.2N$$

Ans

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Problem: 15.40:-

3) A 1.50m long rope is stretched between two Supports with a tension that makes the speed of a transverse waves $48.0ms^{-1}$. What are the wavelength and frequency of the (a) fundamental?

(b) The Second overtone?

(c) The fourth harmonic?

=) In given:-

$L = 1.50m$

$V = 48.0ms^{-1}$

(a)

We know:-

For fundamental $n = 1$

and wavelength $\lambda = \frac{2L}{n}$

$= \frac{2 \times 1.5}{1}$

$\boxed{\lambda = 3m}$ Ans:-

$f = \frac{v}{\lambda}$

$= \frac{48}{3}$

$\boxed{f = 16 Hz.}$ Ans:-

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(b)

We know-
now

2nd Overtone means third harmonic:

for third harmonic $n = 3$

$$\lambda = \frac{2L}{n}$$

$$\therefore \lambda = \frac{2 \times 1.5}{3}$$

$$\boxed{\lambda = 1 m}$$ Ans:-

$$f = \frac{v}{\lambda}$$

$$= \frac{48}{1}$$

$$\boxed{f = 48 Hz}$$ Ans:-

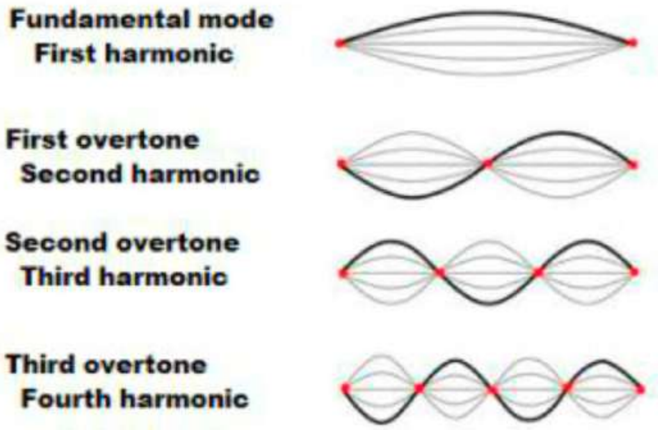

Fig. 1

(c)

We know:-
now

For 4th harmonic $n = 4$

$$\therefore \lambda = \frac{2L}{\lambda} = \frac{2 \times 1.5}{4} = 0.75 m$$

$$f = \frac{v}{\lambda} = \frac{48}{0.75} = 64 Hz$$

Ans:-

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Problem: 15.42:

⇒ A piano tuner stretches a steel piano wire with a tension of 800N. The steel wire is 0.400m long and has a mass 3.00g

(a) What is the frequency of its fundamental modes of vibration?

(b) what is the number of the highest harmonic that could be heard by a person who is capable of hearing frequencies up to 10,000 Hz?

⇒ In given:—

$$T = 800N$$
$$L = 0.400m$$
$$m = 3.00g = 3\times10^{-3} kg$$

(a)

We know:—

For fundamental mode: $n = 1$

$$\therefore \lambda = \frac{2L}{n}$$
$$= \frac{2\times0.400}{1}$$
$$= 0.8m$$

∴

P.T.O

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we know:-

$$V = \sqrt{\frac{T}{\mu}} \quad \text{———} \quad (1)$$

$$\mu = \frac{\Delta m}{\Delta L}$$

$$= \frac{3 \times 10^{-3}}{0.400}$$

$$= 7.5 \times 10^{-3} \, kgm^{-1}$$

Substitute the value of μ in eq (1) =>

$$\therefore V = \sqrt{\frac{800}{7.5 \times 10^{-3}}}$$

$$= 326.59 \, ms^{-1}$$

Now, we know:-

$$V = f\lambda$$

$$\therefore f = \frac{V}{\lambda}$$

$$= \frac{326.59}{0.8}$$

$$= 408.24 \, Hz$$

Ans

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(b)

In given:-

$$f = 10{,}000\ Hz$$

we have:-

$$V = 3.26 \times 10^2\ ms^{-1}$$

$$= 326.52\ ms^{-1}$$

we know:-

$$f = \frac{n}{2L} \times V$$

$$\therefore n = \frac{f2L}{V}$$

$$= \frac{10{,}000 \times 2 \times 0.400}{326.52}$$

$$= 24.5$$

$$\approx 24$$

$\therefore$ The 24th harmonic is the highest that could be heard.

Ans:-